



US012415457B2

(12) **United States Patent**
Maiorana

(10) **Patent No.:** **US 12,415,457 B2**

(45) **Date of Patent:** **Sep. 16, 2025**

(54) **SAFETY ZONE WITH BREAKAWAY FOR
USE ON A TANKER TRUCK**

(56) **References Cited**

U.S. PATENT DOCUMENTS

(71) Applicant: **Mac LTT, Inc.**, Kent, OH (US)

(72) Inventor: **James A. Maiorana**, Hartville, OH
(US)

(73) Assignee: **Mac LTT, Inc.**, Kent, OH (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 1133 days.

(21) Appl. No.: **17/144,508**

(22) Filed: **Jan. 8, 2021**

(65) **Prior Publication Data**

US 2021/0323464 A1 Oct. 21, 2021

Related U.S. Application Data

(60) Provisional application No. 63/013,255, filed on Apr.
21, 2020.

(51) **Int. Cl.**
B60Q 1/30 (2006.01)
B60P 3/22 (2006.01)
B60Q 1/04 (2006.01)
G08G 1/16 (2006.01)

(52) **U.S. Cl.**
CPC **B60Q 1/305** (2013.01); **B60P 3/2245**
(2013.01); **B60P 3/2265** (2013.01); **B60P**
3/228 (2013.01); **G08G 1/168** (2013.01);
B60Q 1/0491 (2013.01)

(58) **Field of Classification Search**
CPC B60Q 1/305; B60P 3/2245; B60P 3/228;
G08G 1/168

See application file for complete search history.

1,365,110 A	1/1921	Kramer	
2,989,213 A	6/1961	Daggit	
3,788,268 A	1/1974	Hiatt et al.	
4,311,208 A	1/1982	Macrorie et al.	
4,371,056 A	2/1983	Anglade	
4,666,176 A *	5/1987	Sand	B60D 1/36 280/477
4,679,657 A	7/1987	Bennett et al.	
4,697,541 A	10/1987	Wicker	
4,825,192 A	4/1989	Wells	
4,848,517 A	7/1989	Broyles	
5,113,588 A *	5/1992	Walston	B60D 1/36 280/477
5,497,304 A	3/1996	Caine	
5,847,642 A	12/1998	Esposito et al.	
5,878,676 A	3/1999	Etling et al.	
5,878,767 A	3/1999	Etling et al.	
5,907,284 A	5/1999	Israel	
5,921,199 A *	7/1999	Gross	B60Q 1/482 116/28 R
5,971,105 A	10/1999	Jacobson	
6,092,911 A	7/2000	Baker, III	

(Continued)

FOREIGN PATENT DOCUMENTS

EP 2902565 8/2015

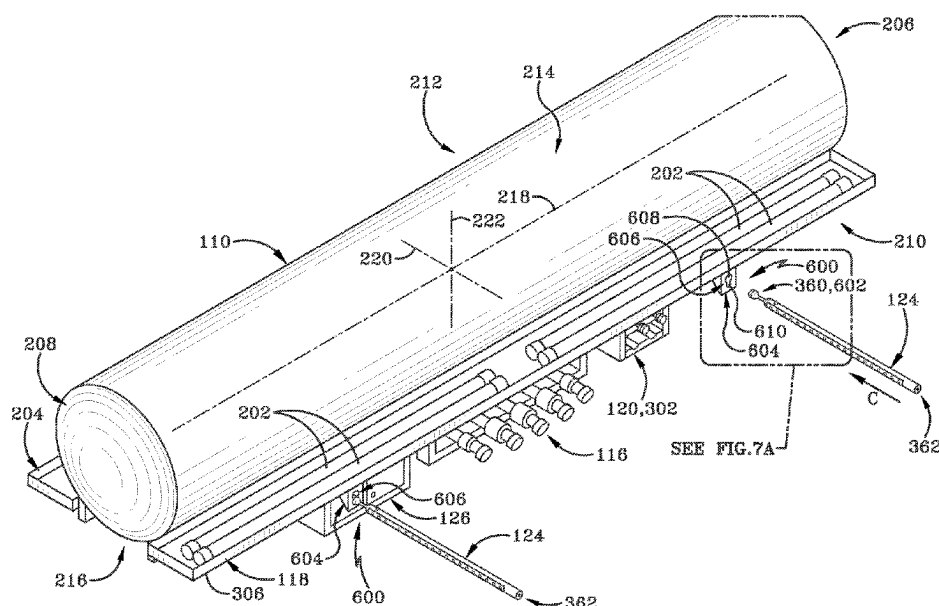
Primary Examiner — Kevin Hurley

(74) Attorney, Agent, or Firm — Sand, Sebolt & Wernow
Co., LPA

(57) **ABSTRACT**

A vehicle includes a safety arm and a coupler on the safety
arm. The coupler couples the first safety arm to the vehicle.
The first safety arm is adapted to fall away from the vehicle
without damaging the vehicle when the safety arm contacts
an object.

25 Claims, 28 Drawing Sheets



(56)	References Cited		2005/0173189 A1	8/2005	Berardi	
	U.S. PATENT DOCUMENTS		2005/0194761 A1 *	9/2005	Givens	B60D 1/36 280/477
	6,118,372 A	9/2000	Leow	2008/0179908 A1	7/2008	Haber et al.
	6,213,047 B1	4/2001	Means et al.	2010/0018800 A1	1/2010	Ventura et al.
	6,302,567 B1 *	10/2001	Gamble, Sr. B60Q 1/2657 362/540	2010/0118557 A1	5/2010	Ortwein et al.
	6,431,093 B1	8/2002	Hansen	2010/0224444 A1	9/2010	Simeonov et al.
	6,585,281 B1 *	7/2003	Voorting B60D 1/36 280/477	2011/0148645 A1	6/2011	Samiento
	6,739,349 B2	5/2004	Kastenschmidt et al.	2011/0202161 A1	8/2011	Krieger et al.
	6,785,991 B2	9/2004	Colip	2013/0140111 A1	6/2013	Desai
	6,921,184 B2	7/2005	Tufte	2013/0146494 A1	6/2013	Gilpatrick
	6,966,403 B1	11/2005	Chandra	2013/0235558 A1	9/2013	Zhou
	7,174,994 B1	2/2007	Coffield et al.	2013/0265784 A1	10/2013	Nieberle
	7,370,602 B2	5/2008	Greves	2013/0313978 A1	11/2013	Fields et al.
	7,690,858 B1	4/2010	Chiavola	2014/0041964 A1	2/2014	Bedyk
	7,841,269 B1	11/2010	Jacobs	2014/0056013 A1	2/2014	Hancock, Jr. et al.
	8,075,169 B2	12/2011	Englander	2014/0353081 A1	12/2014	Linch et al.
	8,118,250 B1	2/2012	Helseth et al.	2015/0075908 A1	3/2015	Moss et al.
	8,232,888 B2	7/2012	Frederick et al.	2015/0203026 A1	7/2015	Schotanus
	8,316,910 B2	11/2012	Popa et al.	2015/0241034 A1	8/2015	Dankelmann et al.
	8,371,742 B2	2/2013	Doron et al.	2016/0090028 A1 *	3/2016	Krejci B60Q 1/2657 362/485
	8,552,852 B1	10/2013	Hertz	2016/0230461 A1	8/2016	Finegan
	8,610,598 B2	12/2013	Hider et al.	2016/0273744 A1	9/2016	Stout
	8,616,721 B1	12/2013	Woodall, Jr. et al.	2016/0281885 A1	9/2016	Shearer et al.
	RE44,787 E	3/2014	Burgess	2016/0305619 A1	10/2016	Howe
	8,698,610 B2	4/2014	Krugh, IV	2016/0347247 A1	12/2016	Espey et al.
	8,698,612 B2	4/2014	Toll	2016/0360593 A1	12/2016	Lee
	8,740,267 B1	6/2014	Wilson	2017/0036595 A1	2/2017	Wright
	8,770,778 B2	7/2014	Pereira et al.	2017/0068303 A1	3/2017	Land et al.
	9,006,172 B2	4/2015	Cott, Sr. et al.	2017/0188439 A1	6/2017	Martinez
	9,442,194 B2	9/2016	Kurihara et al.	2017/0232930 A1	8/2017	Murar et al.
	9,649,899 B1 *	5/2017	Berry B60D 1/36	2017/0241203 A1	8/2017	Elnore
	9,694,883 B1	7/2017	Bonventre	2017/0283240 A1	10/2017	Wruck et al.
	9,745,706 B2	8/2017	Baban et al.	2017/0291535 A1	10/2017	Arcand
	10,766,410 B2 *	9/2020	Maiorana G08B 5/36	2017/0314328 A1	11/2017	Friedman
	10,836,306 B1	11/2020	Dubey et al.	2017/0344097 A1	11/2017	Land et al.
	11,155,204 B2 *	10/2021	Maiorana B60Q 1/05	2017/0350584 A1	12/2017	Hulet et al.
	11,343,431 B2	5/2022	Tonkin	2018/0122218 A1	5/2018	Shanley
	11,420,525 B2 *	8/2022	Maiorana B60Q 1/32	2021/0053484 A1	2/2021	Paton
	11,634,095 B2 *	4/2023	Maiorana B60Q 1/247 40/556	2021/0107582 A1	4/2021	Hillman et al.
	2004/0173406 A1	9/2004	Lantz	2021/0129932 A1	5/2021	Keller

* cited by examiner

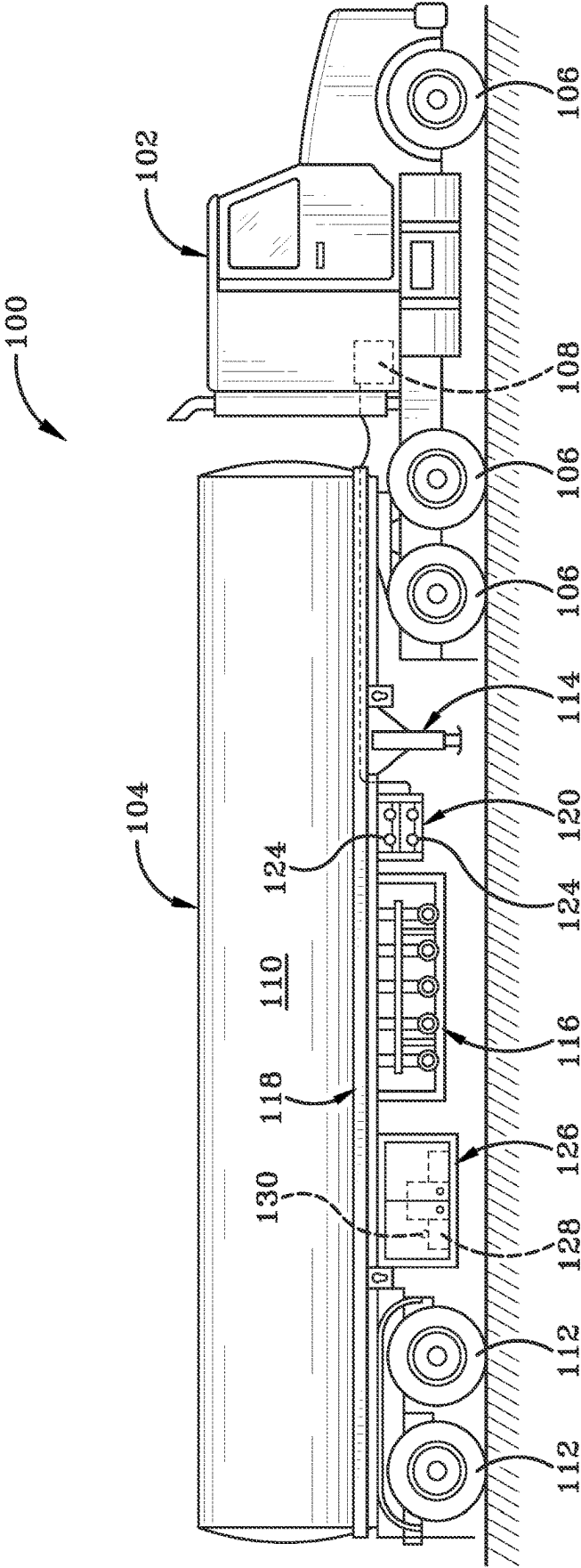
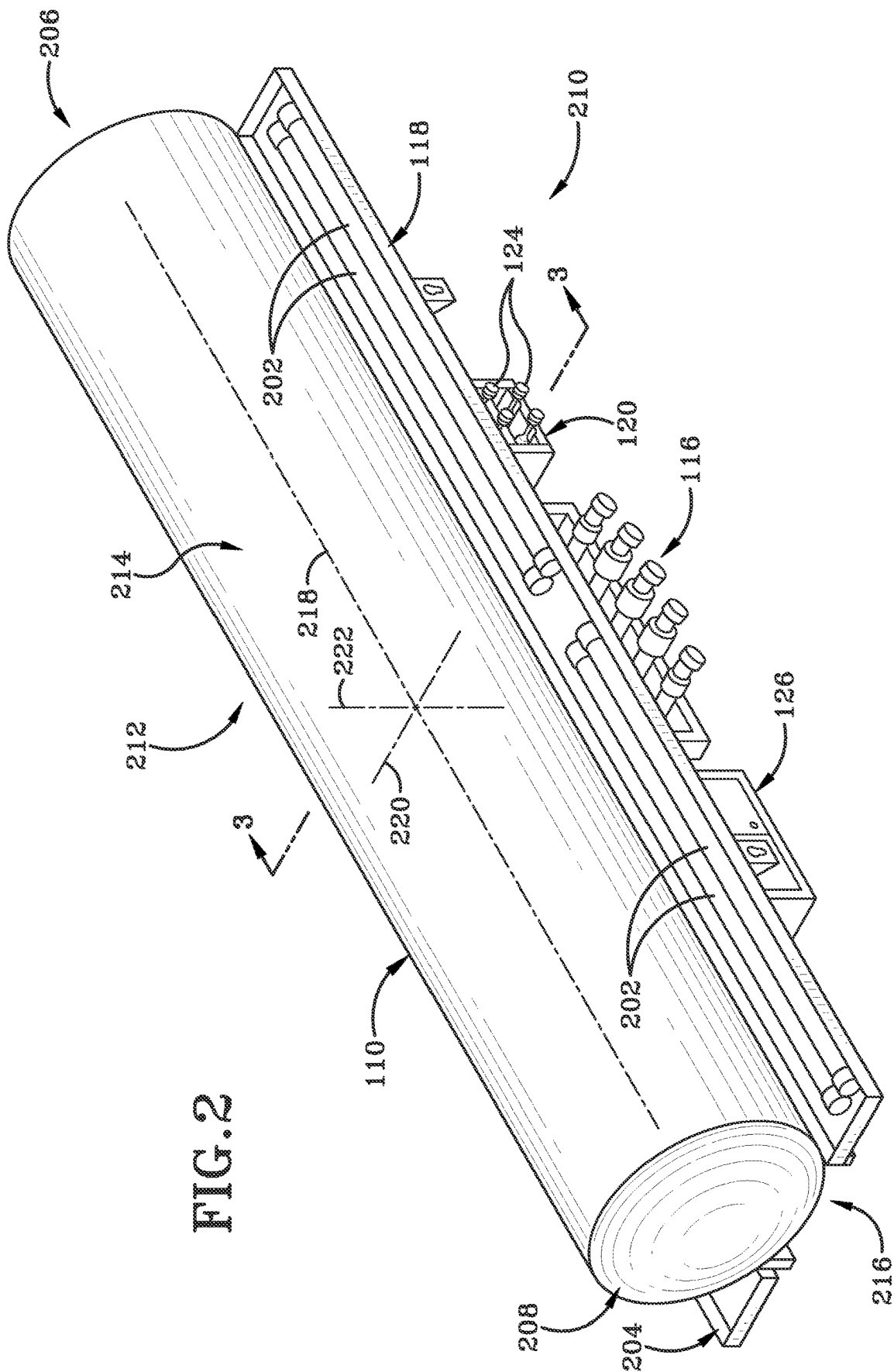
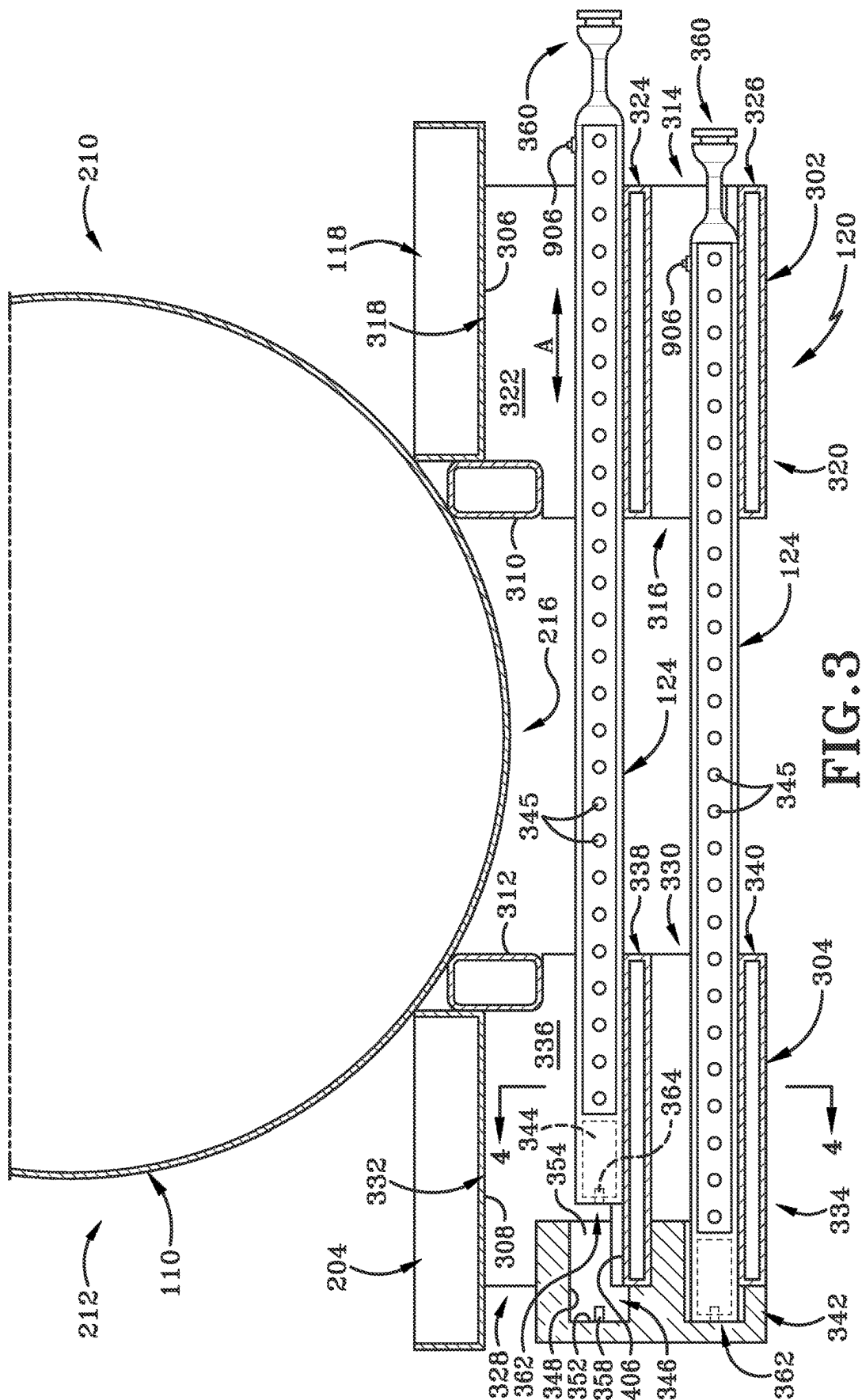
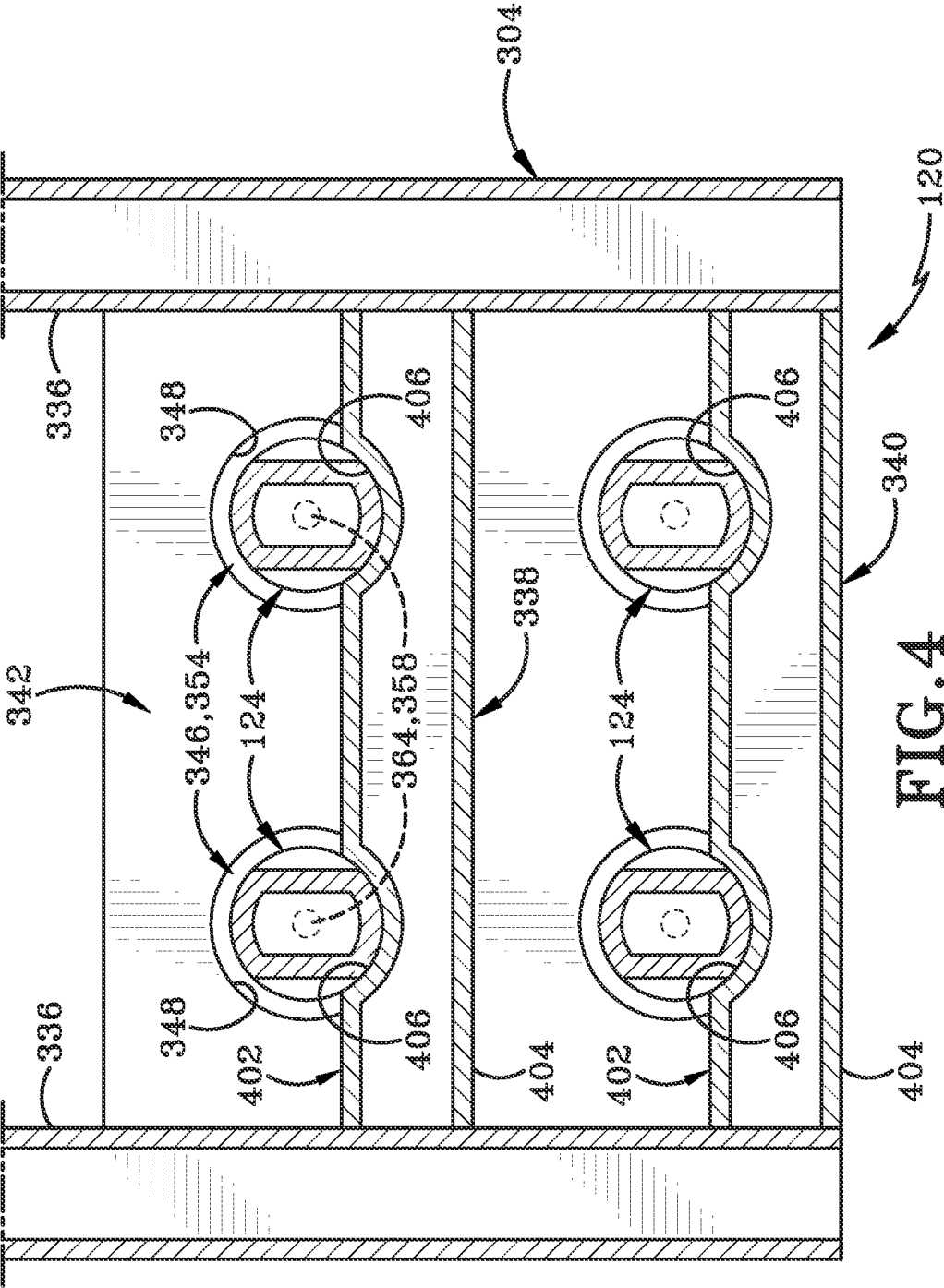
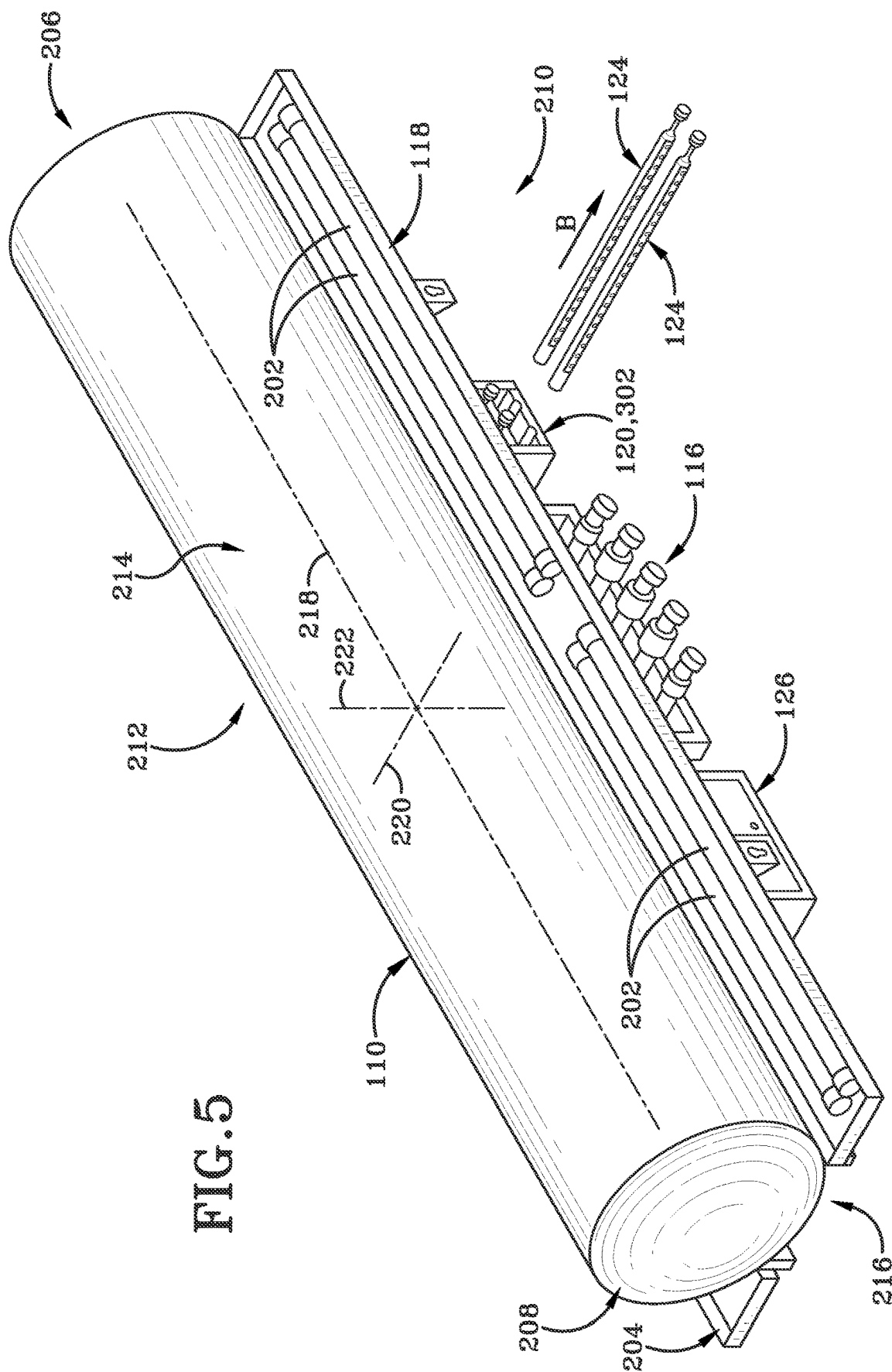


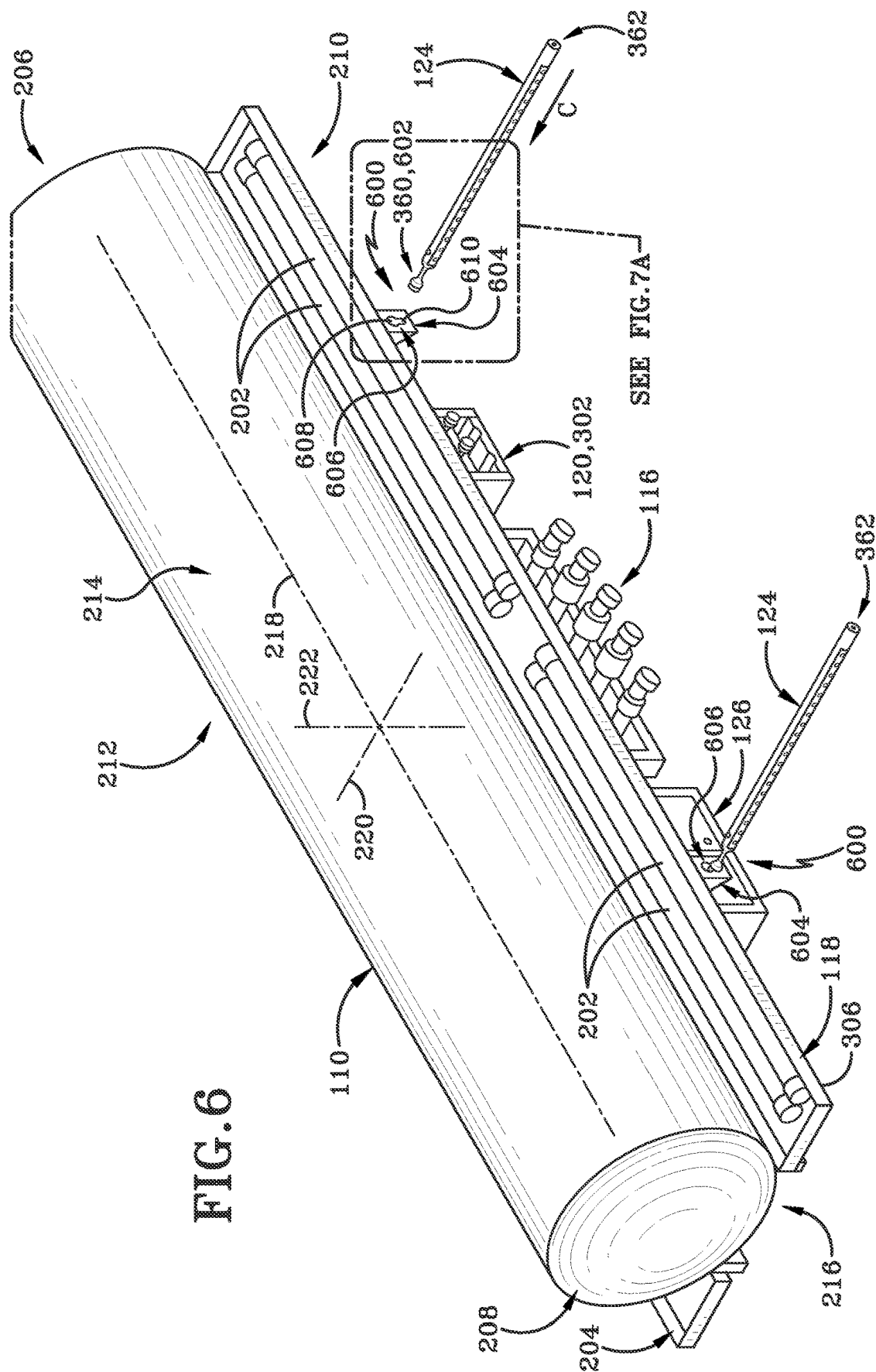
FIG. 1

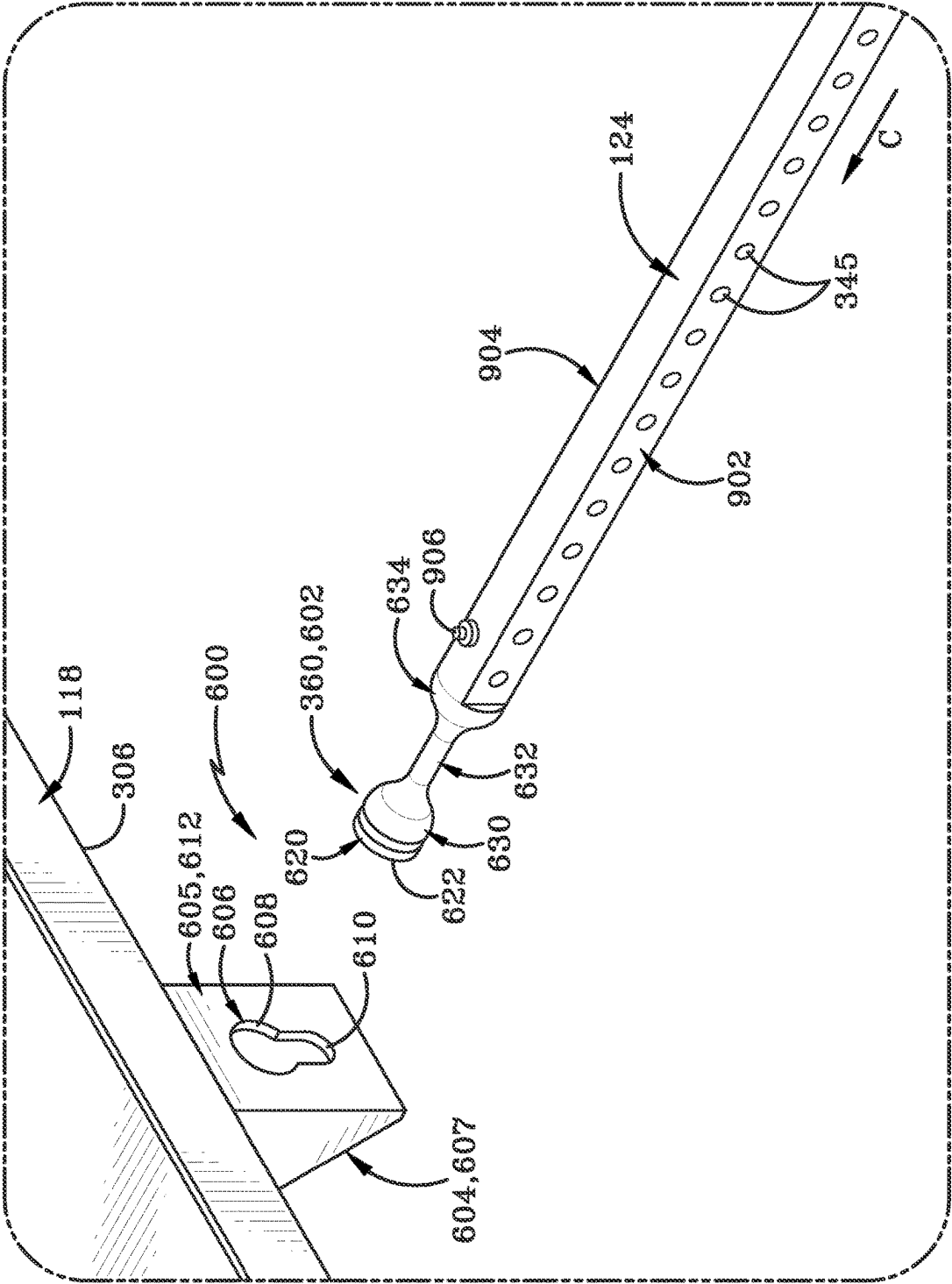












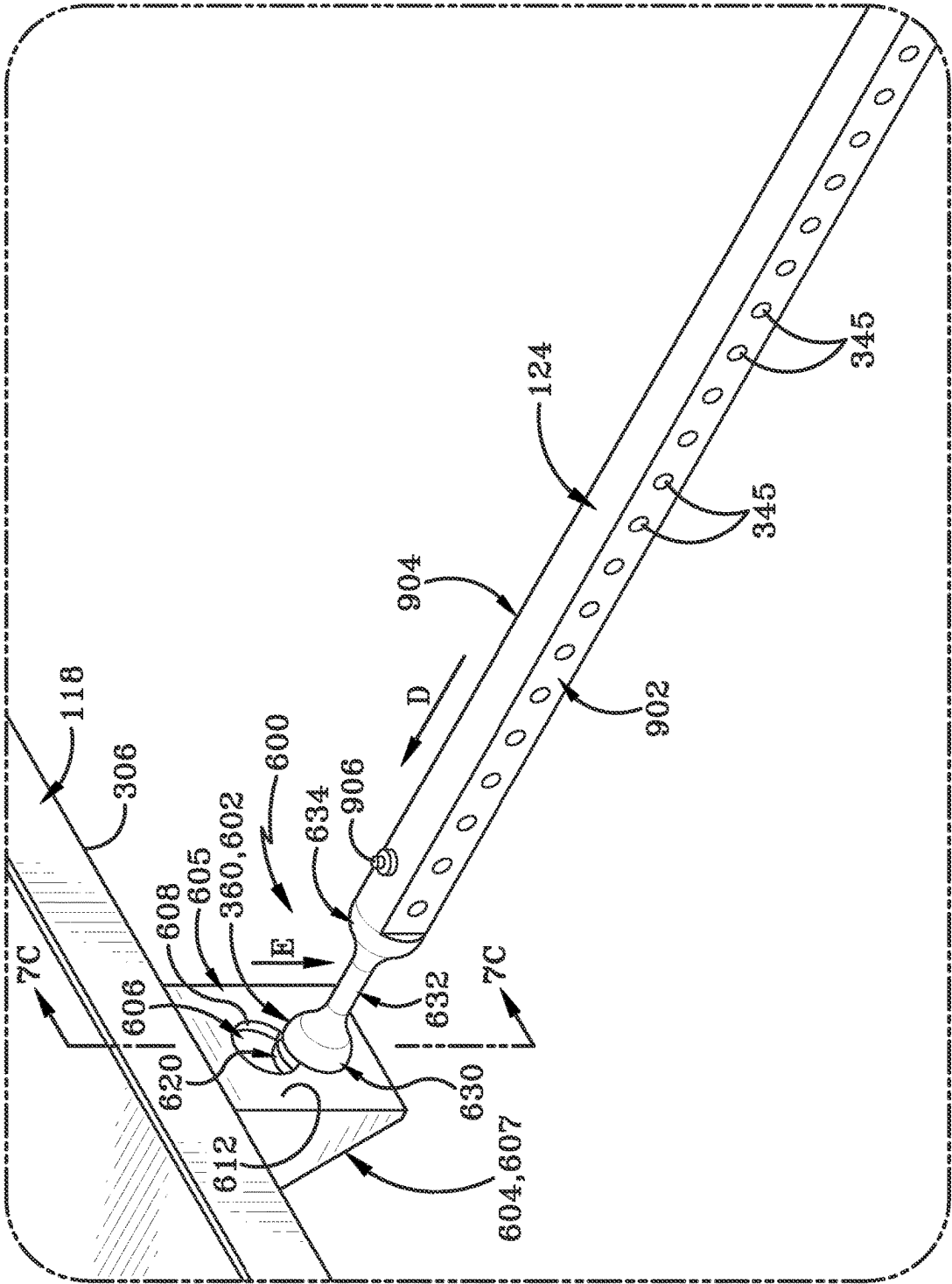
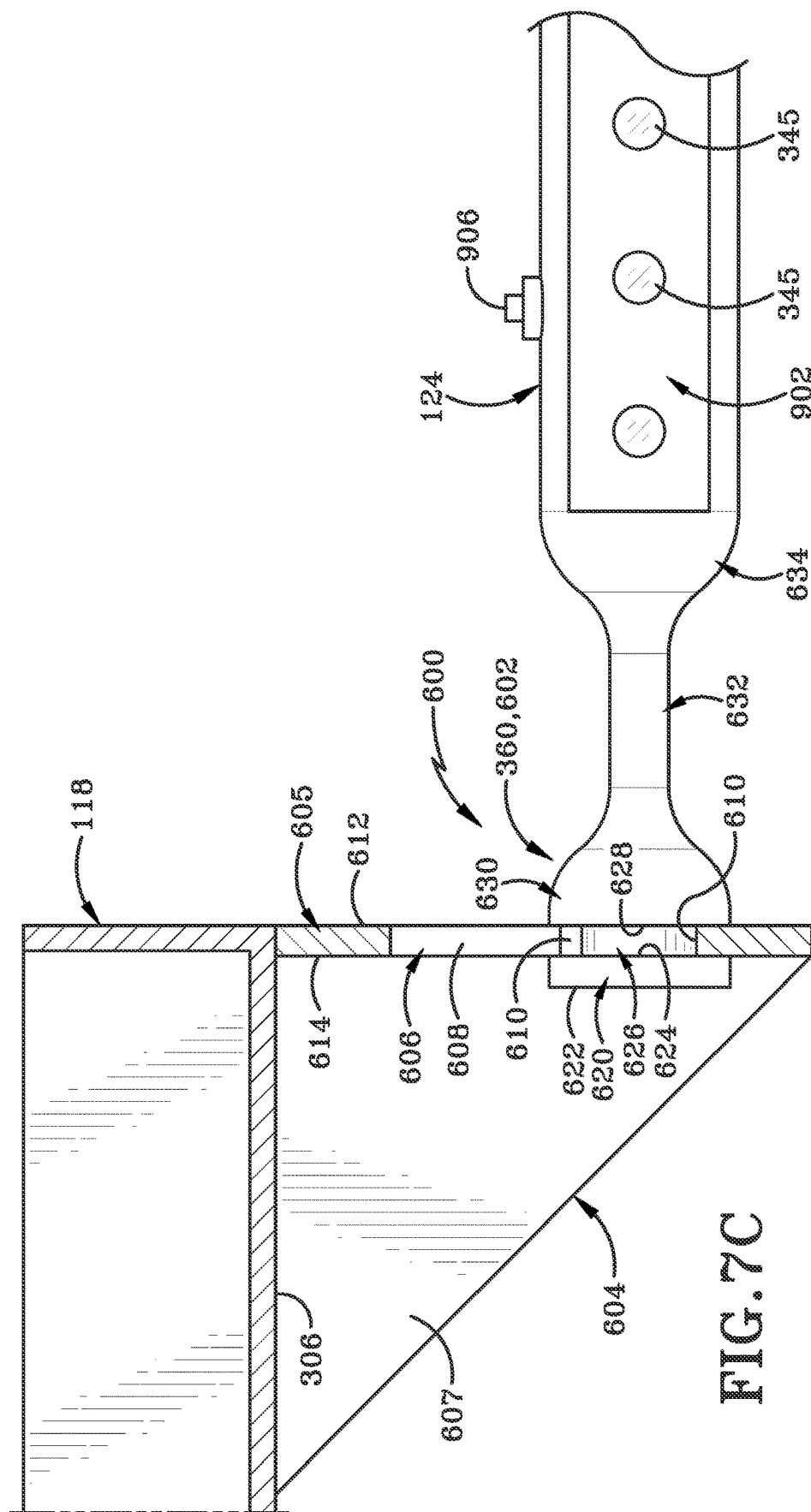


FIG. 7B



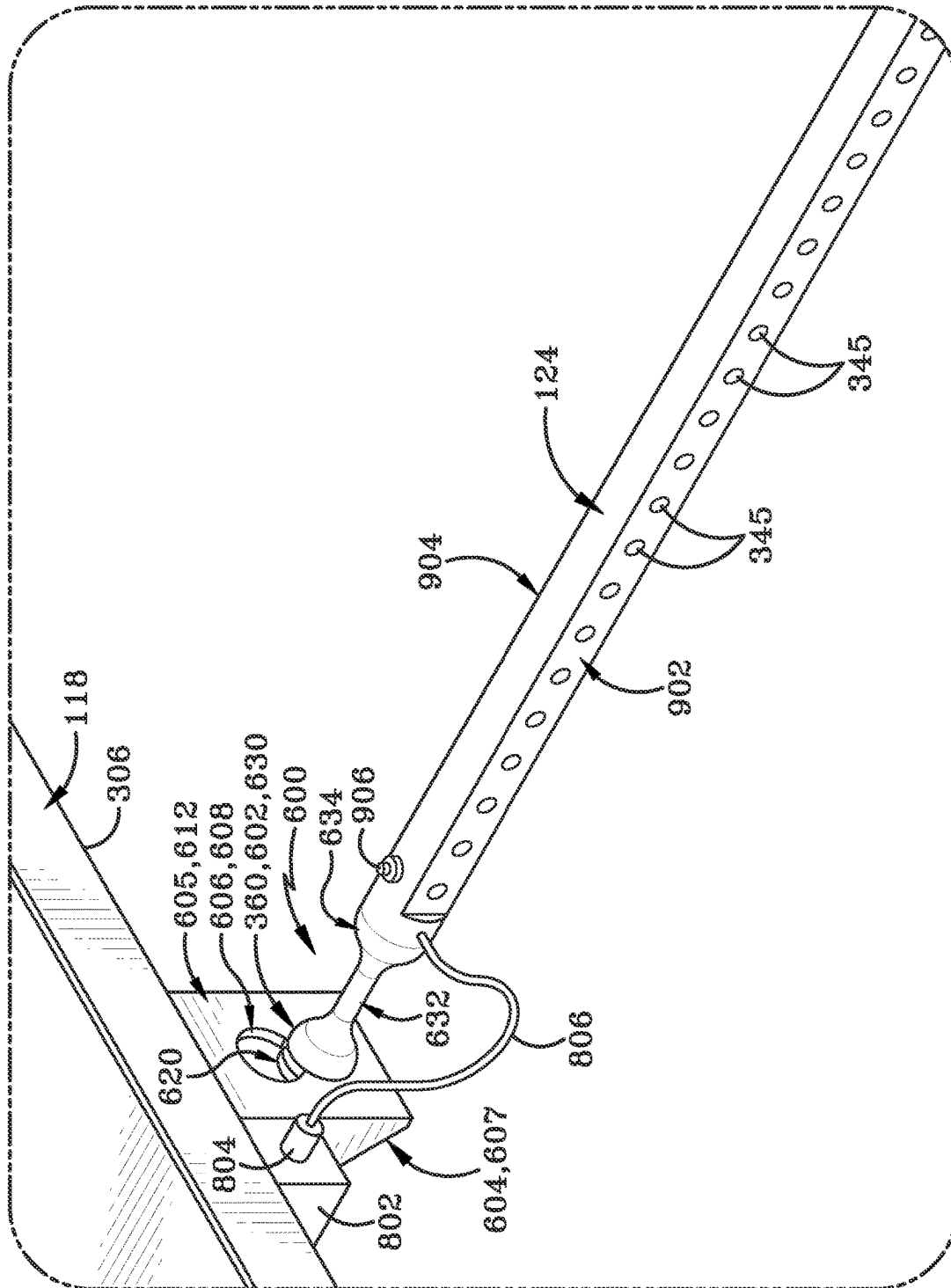
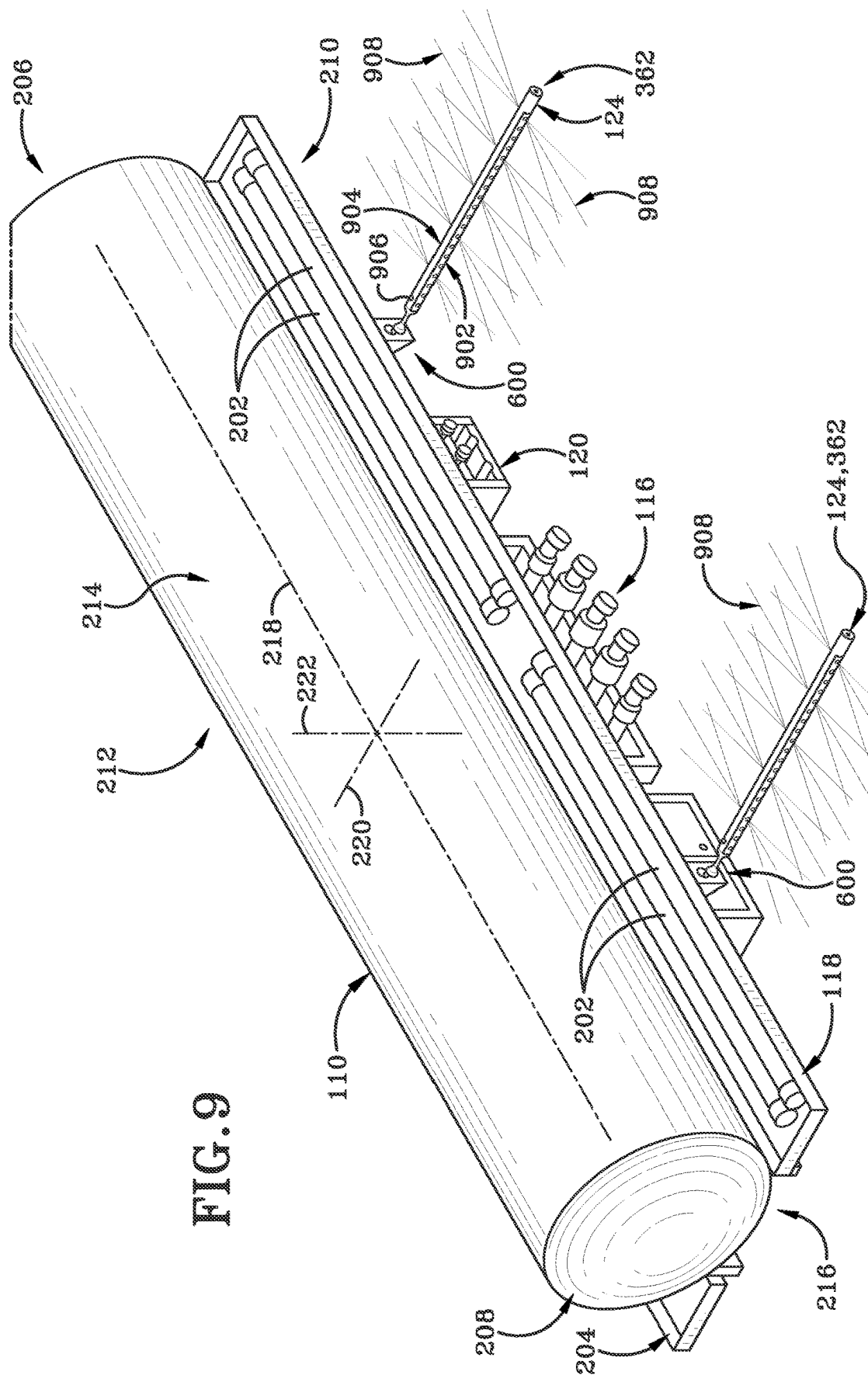
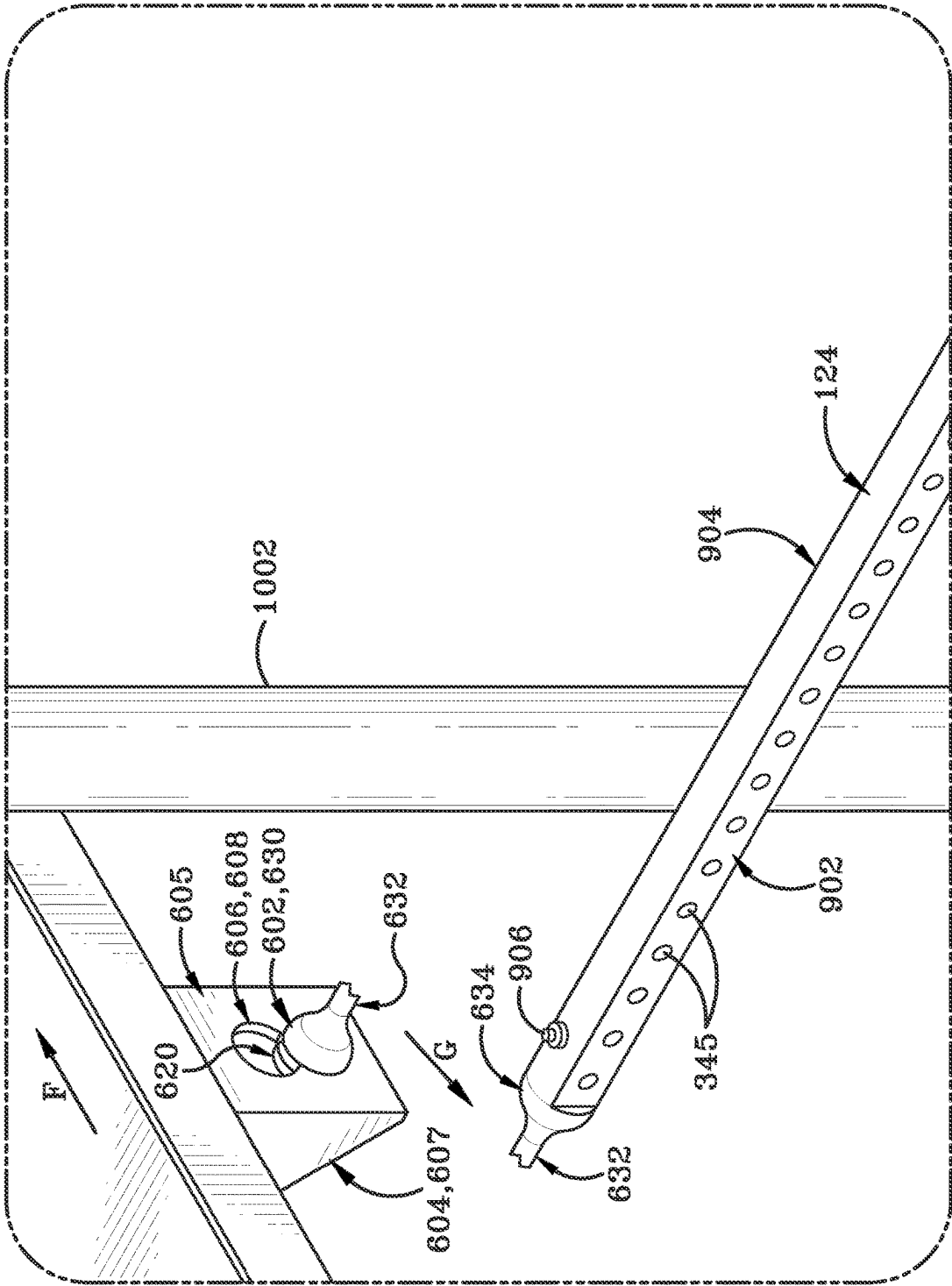


FIG.





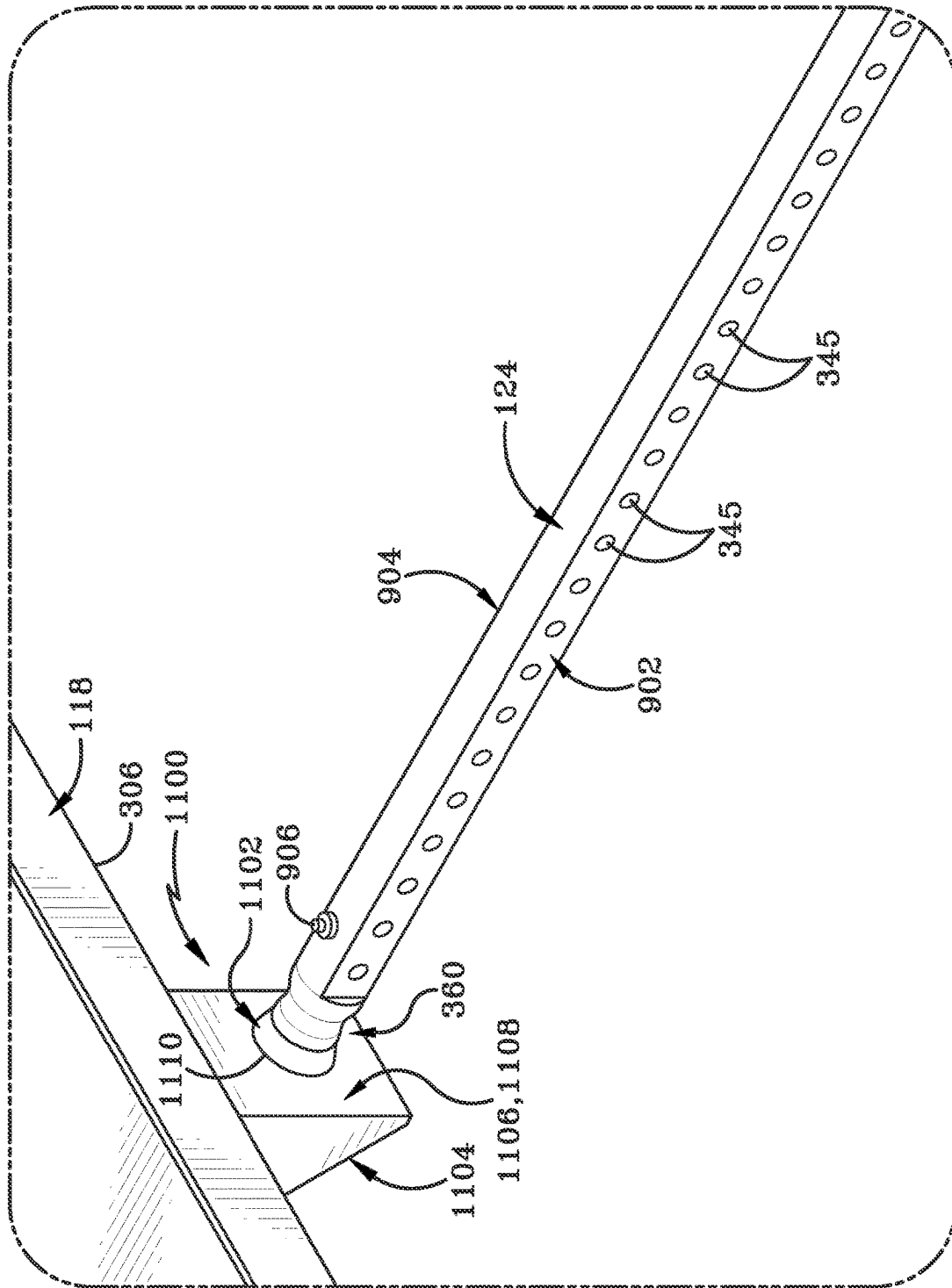


FIG. 11

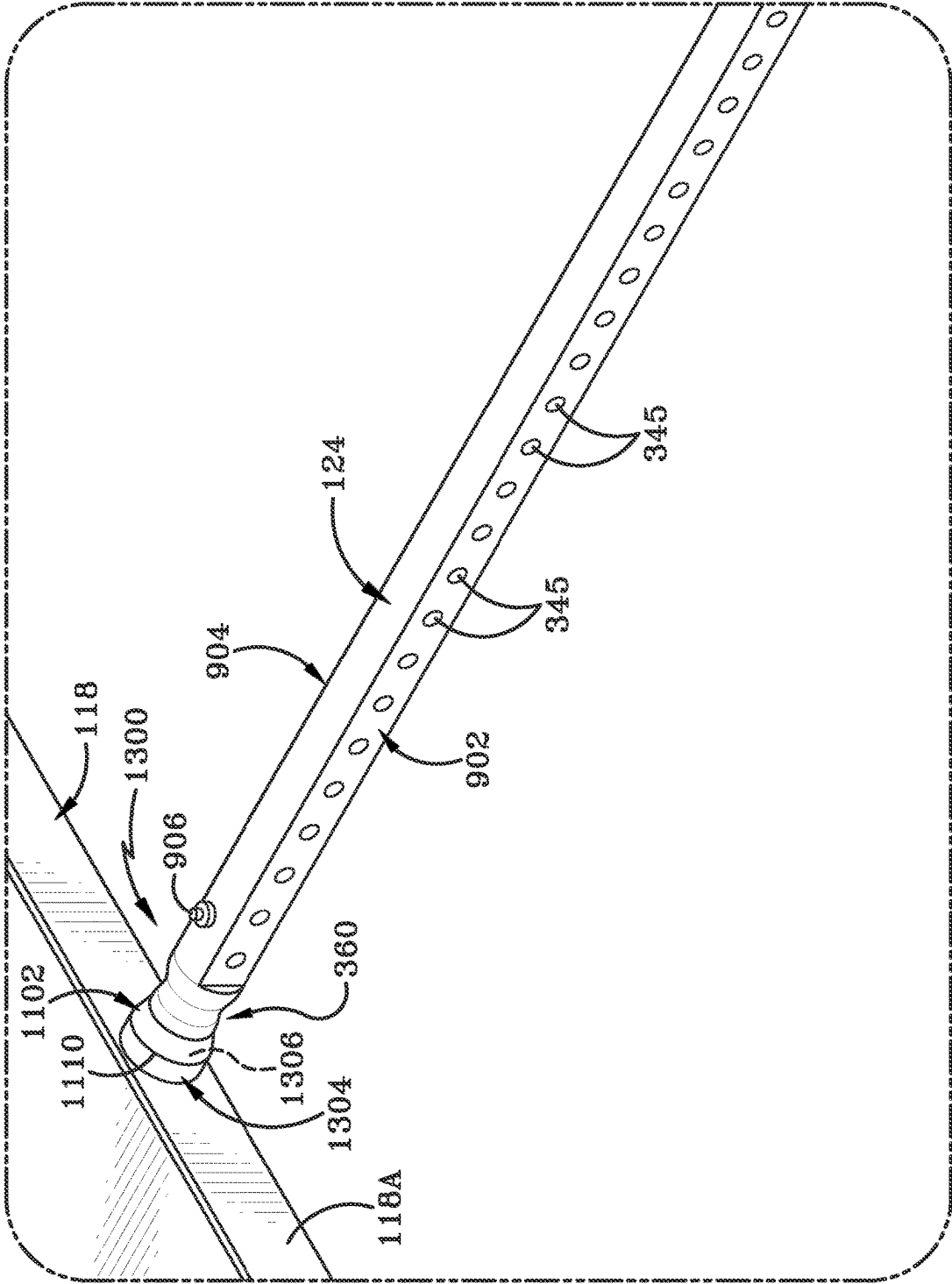


FIG.13

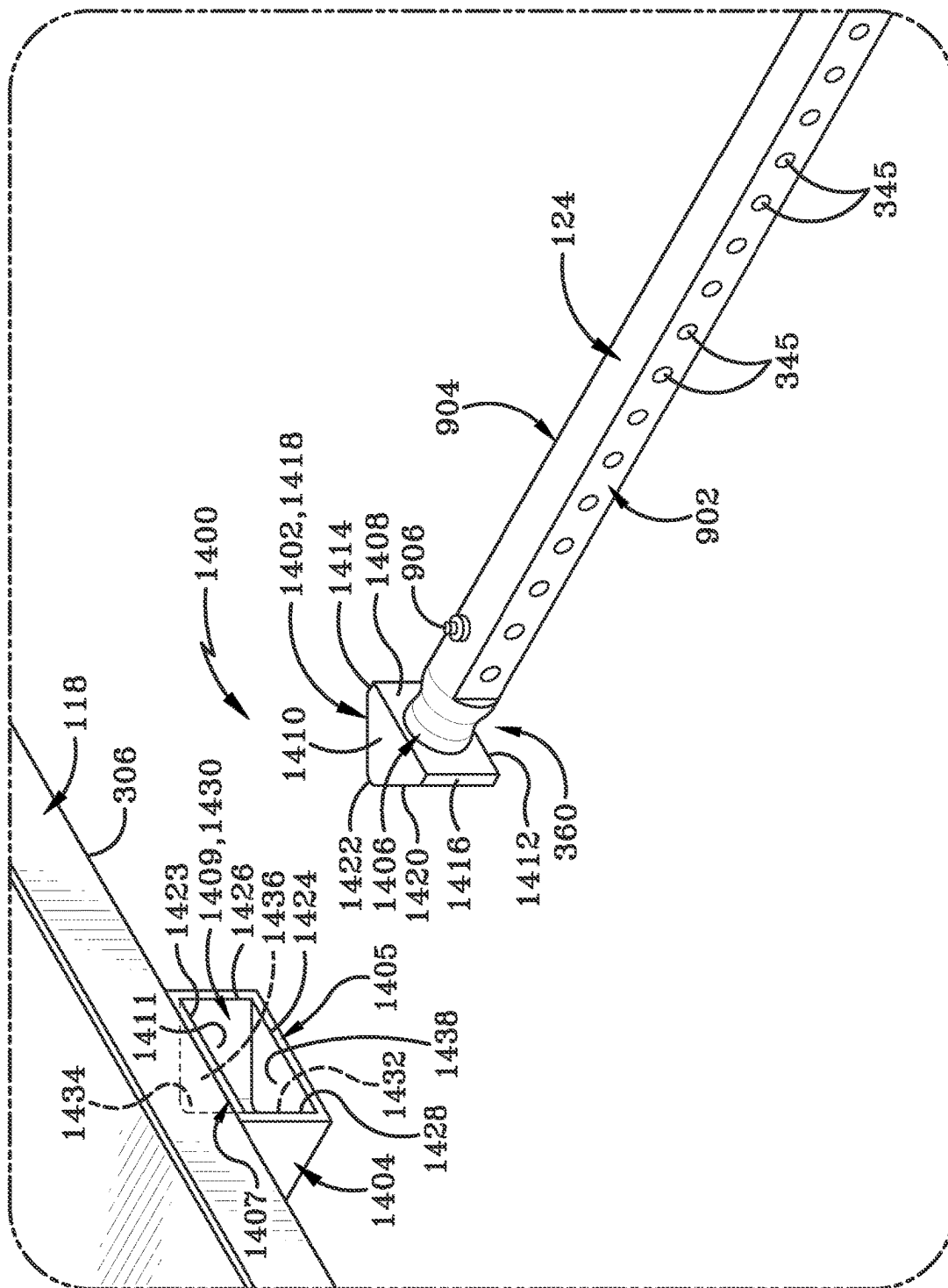


FIG. 14

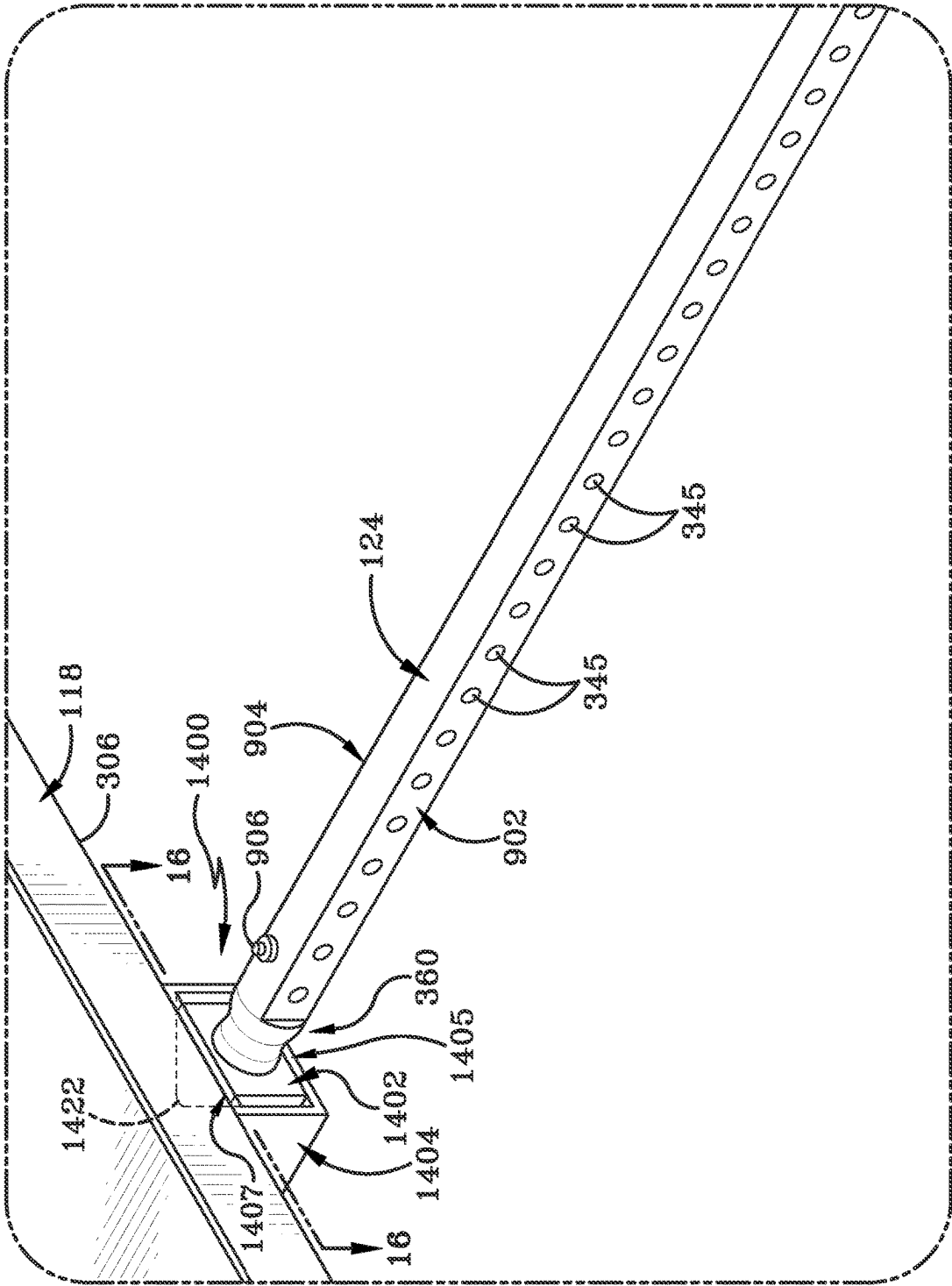


FIG.15

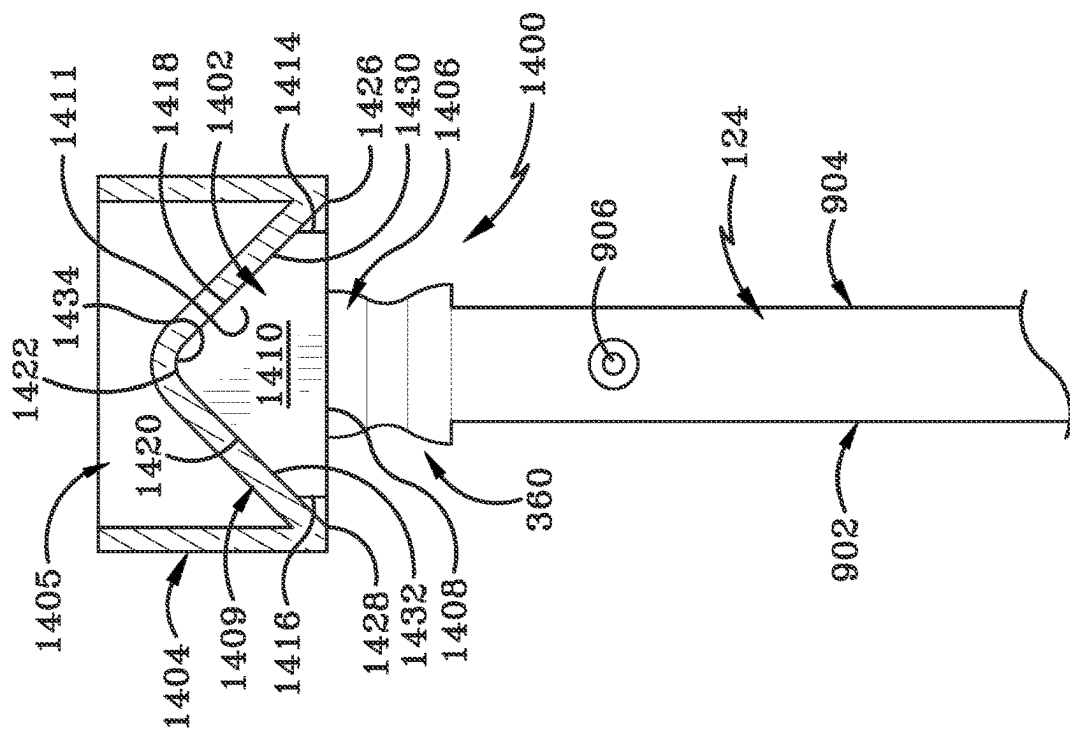


FIG. 16

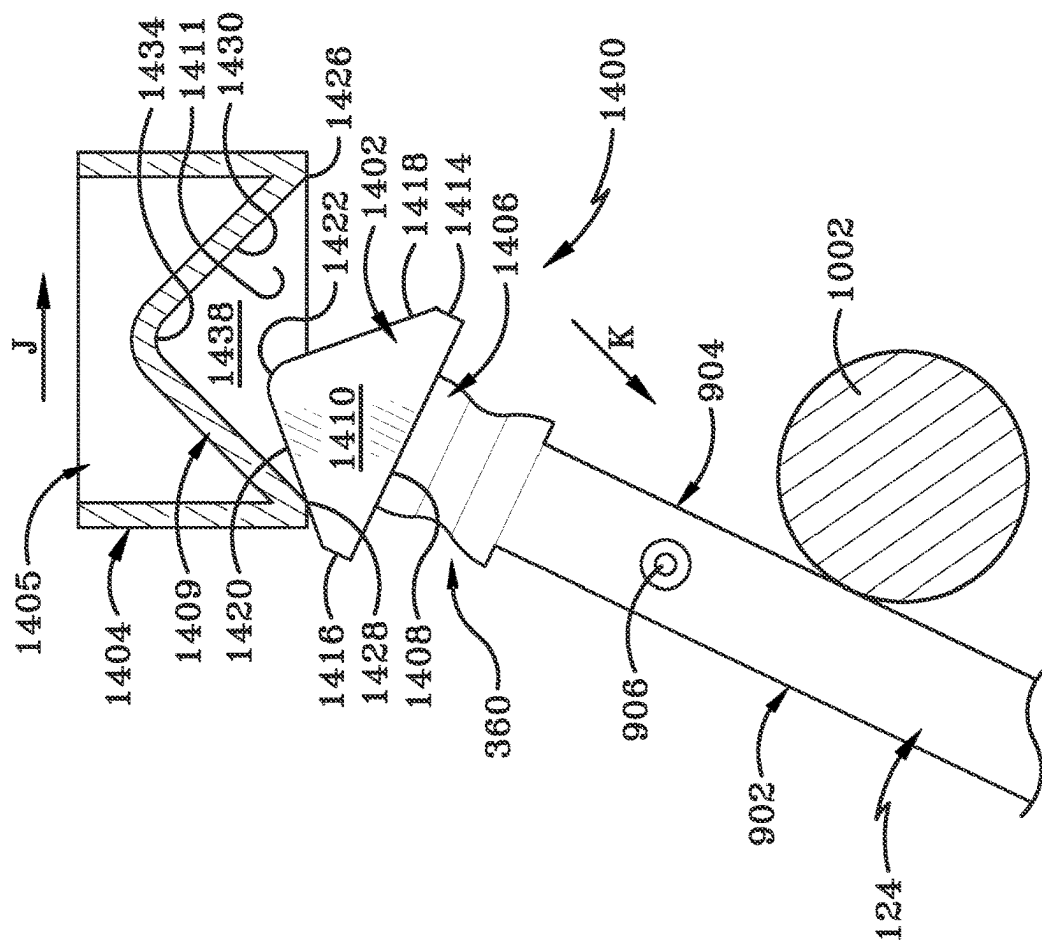
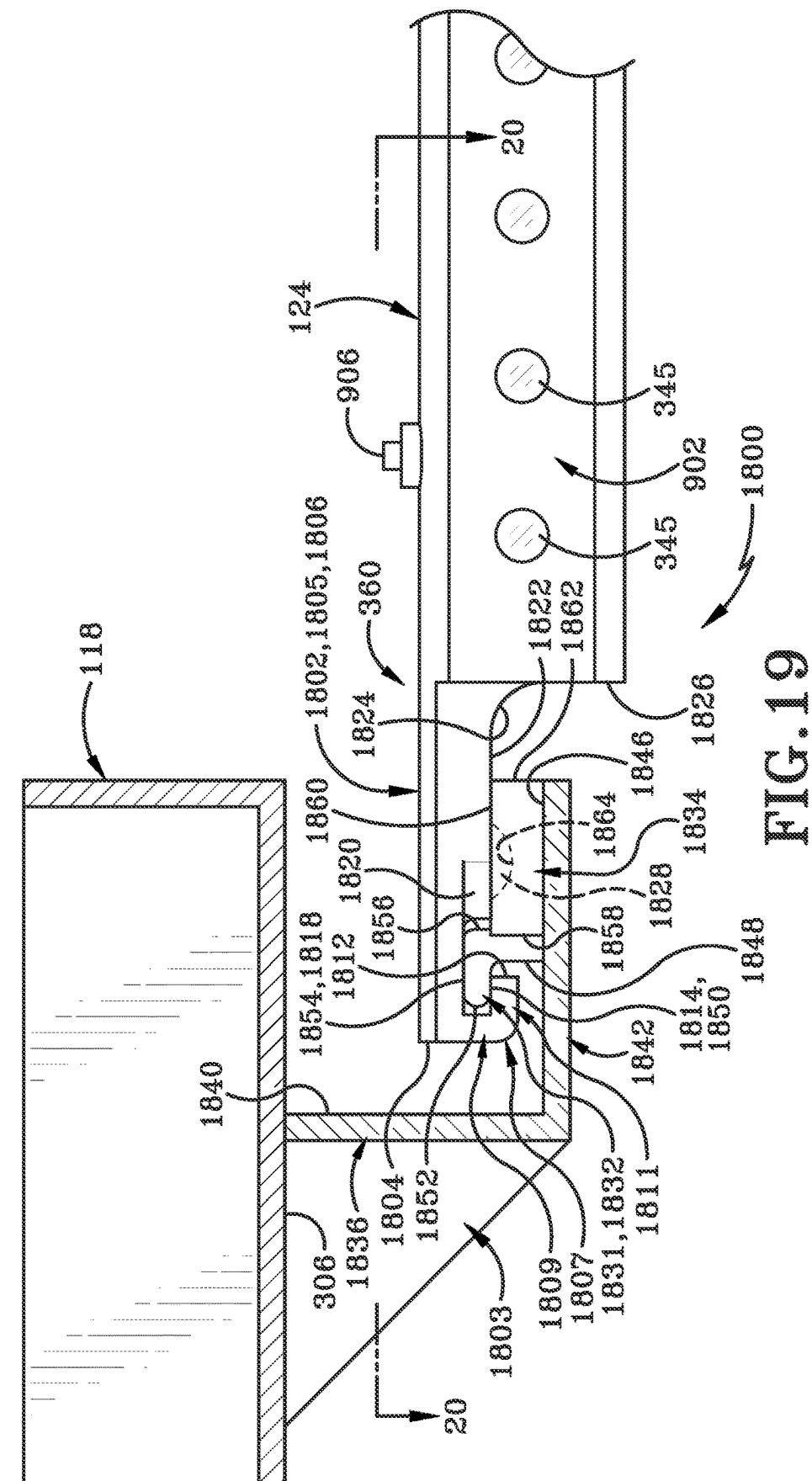


FIG. 17



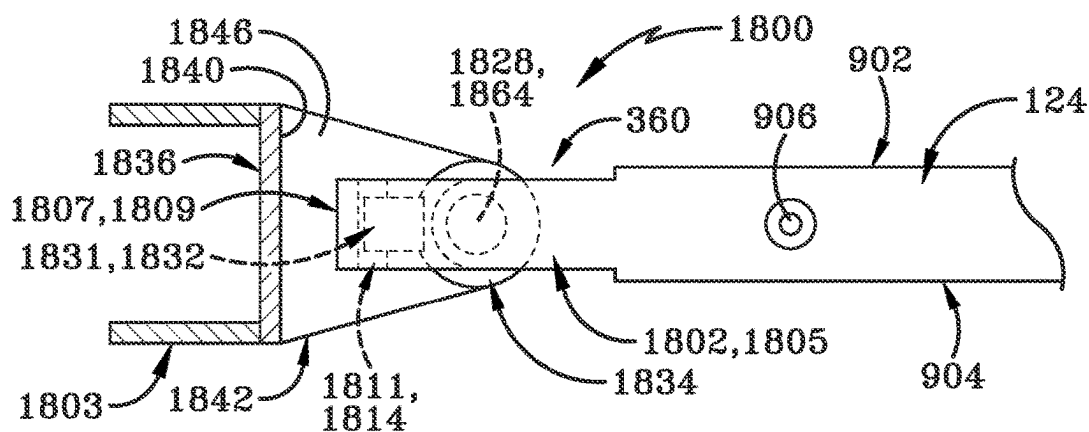


FIG. 20

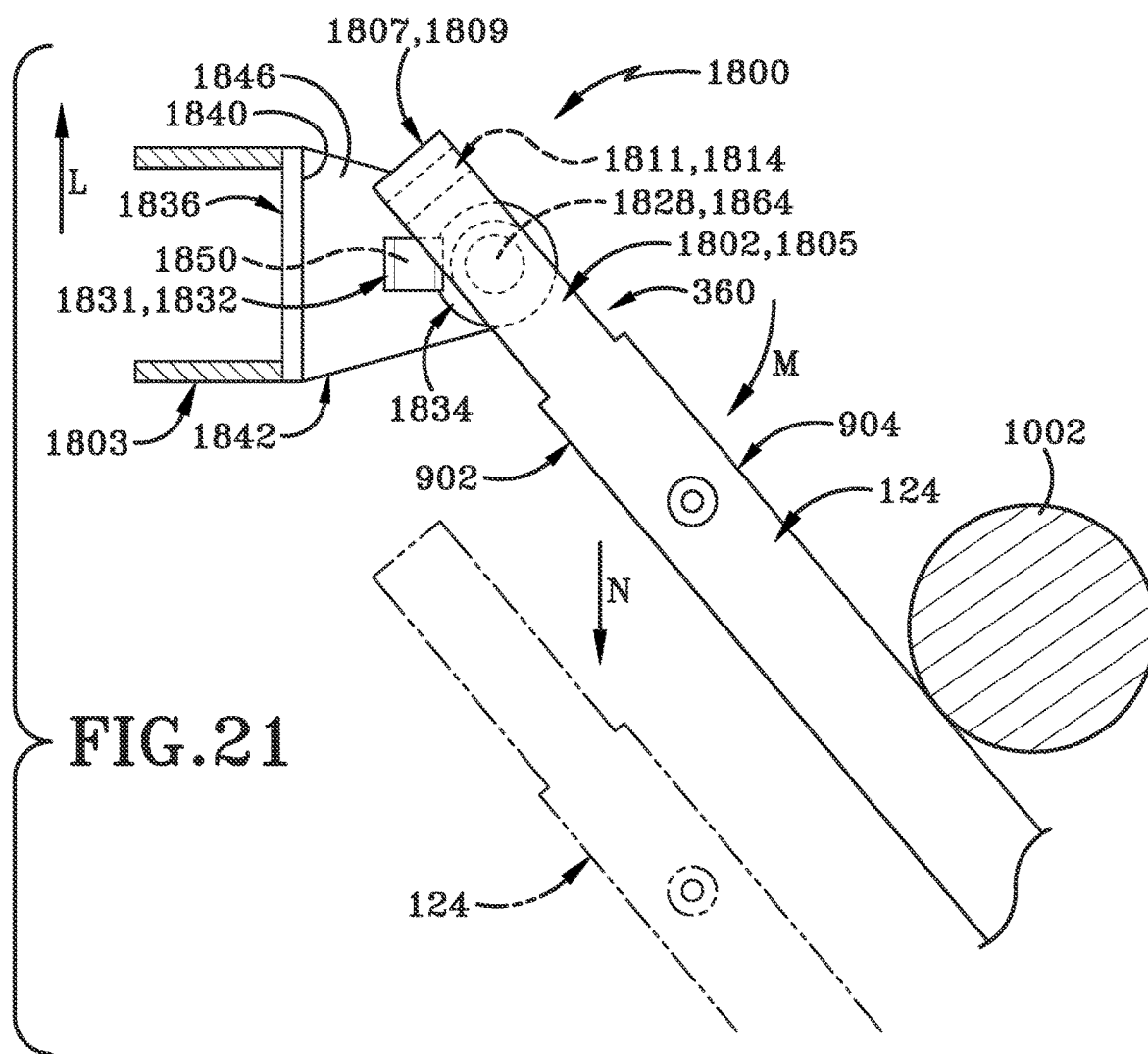
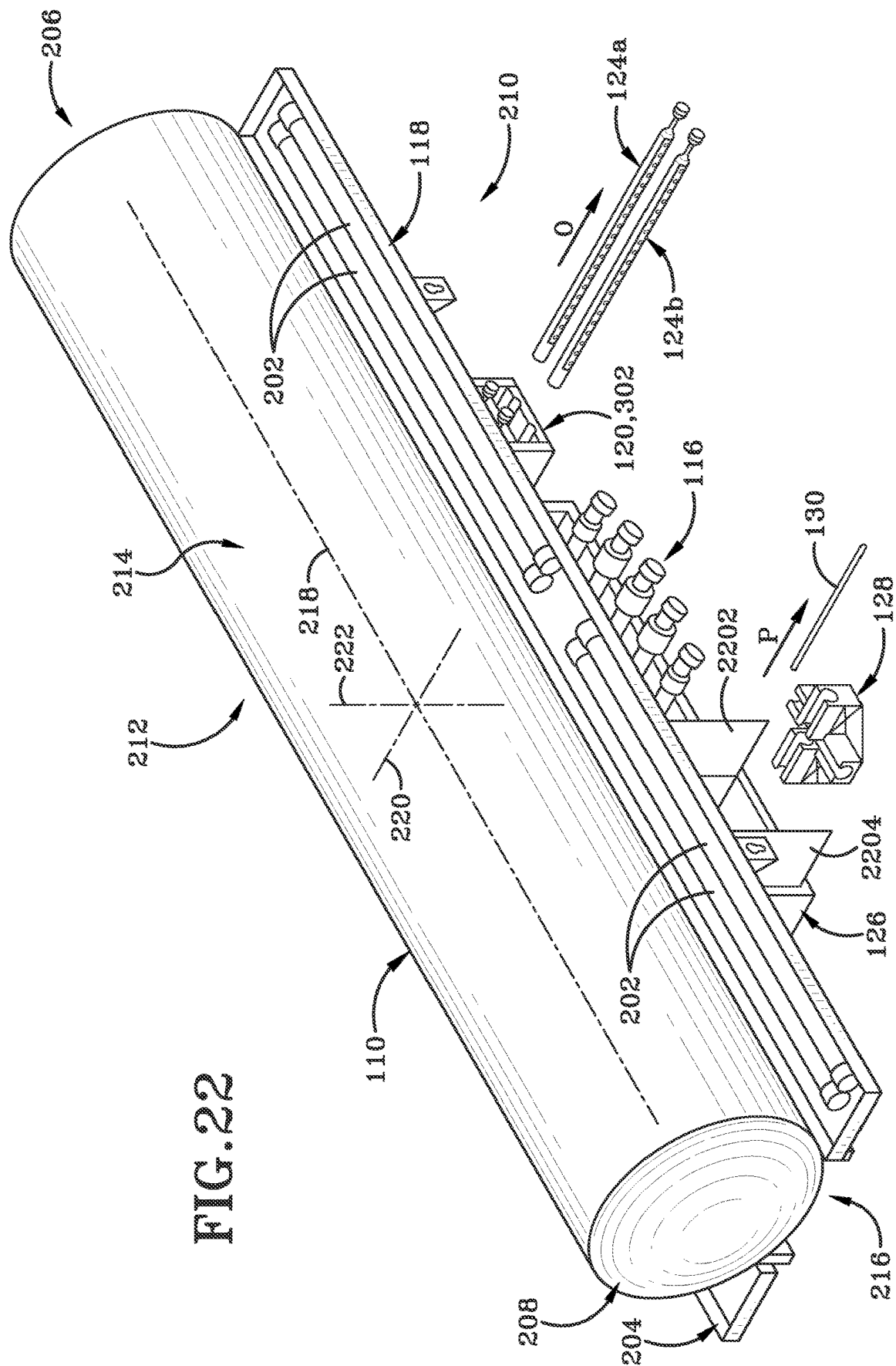


FIG. 21



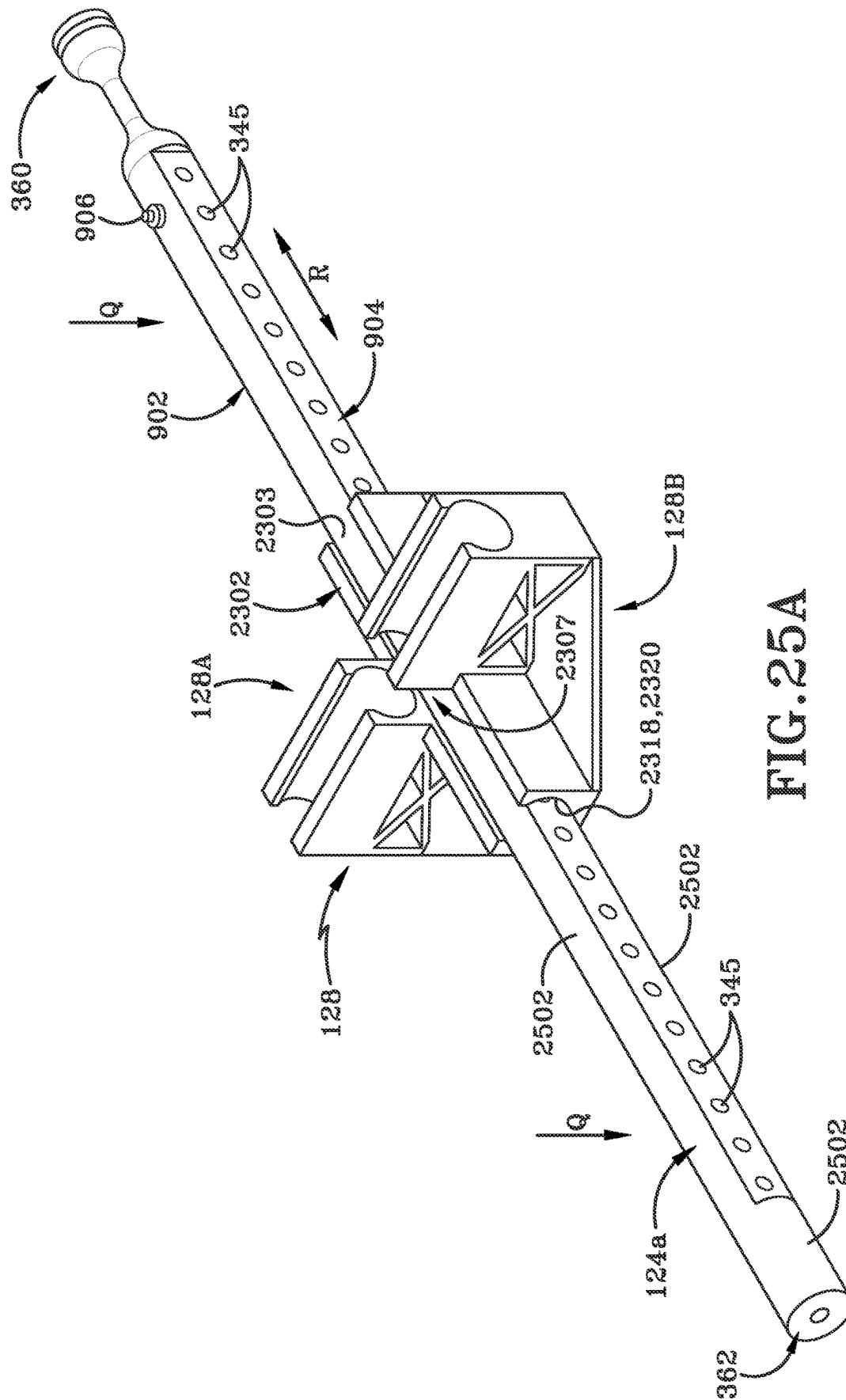
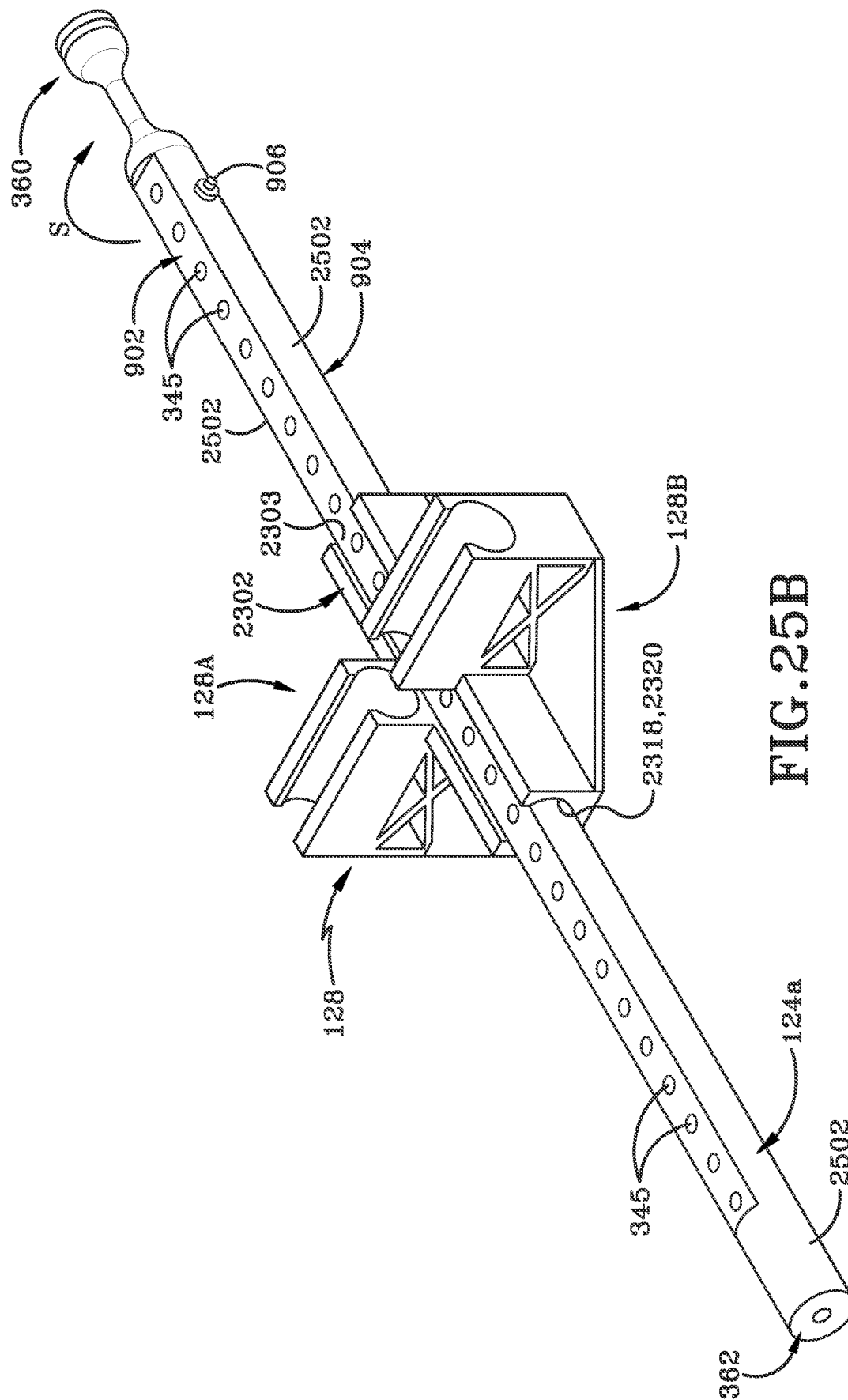


FIG. 25A



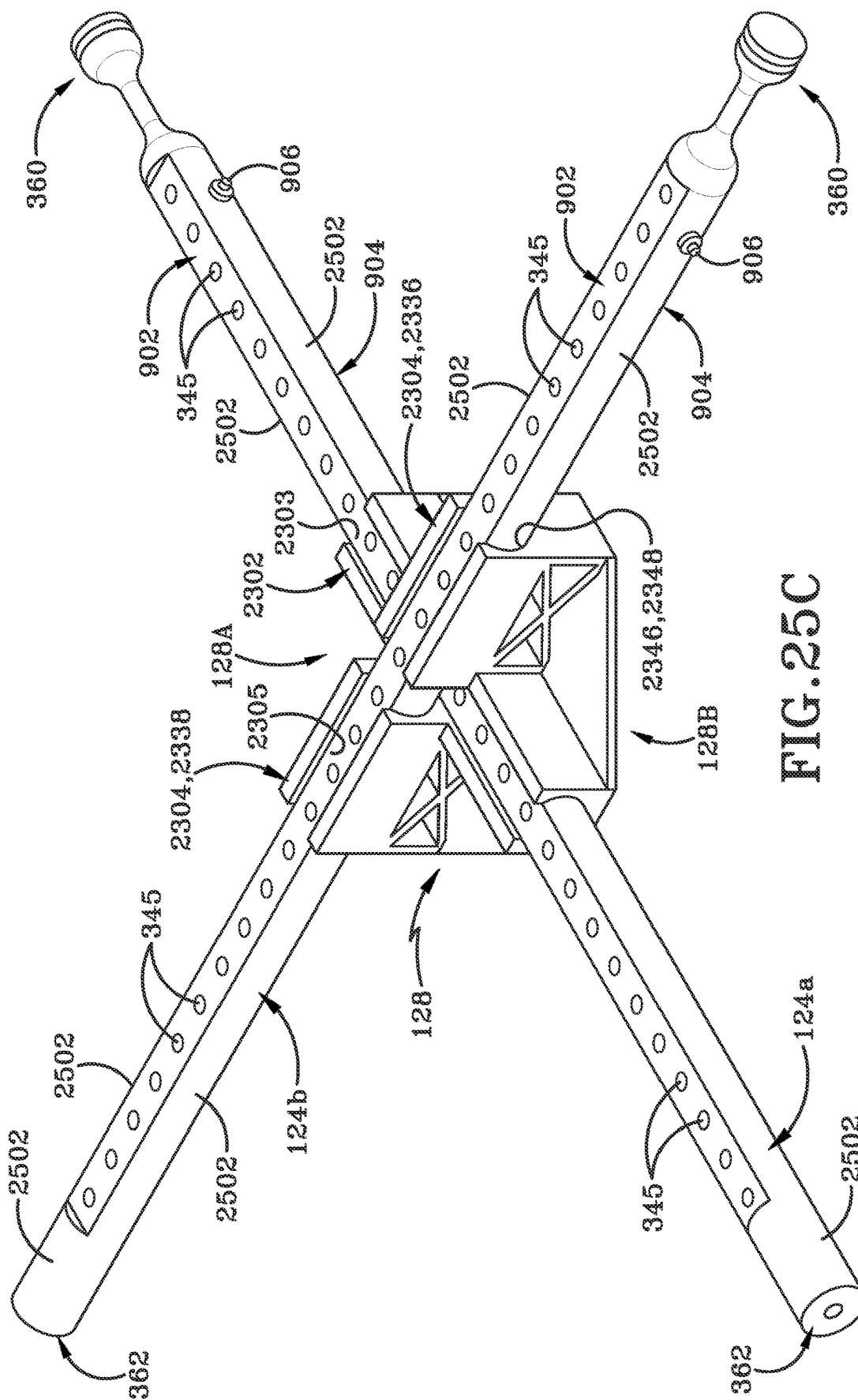


FIG. 25C

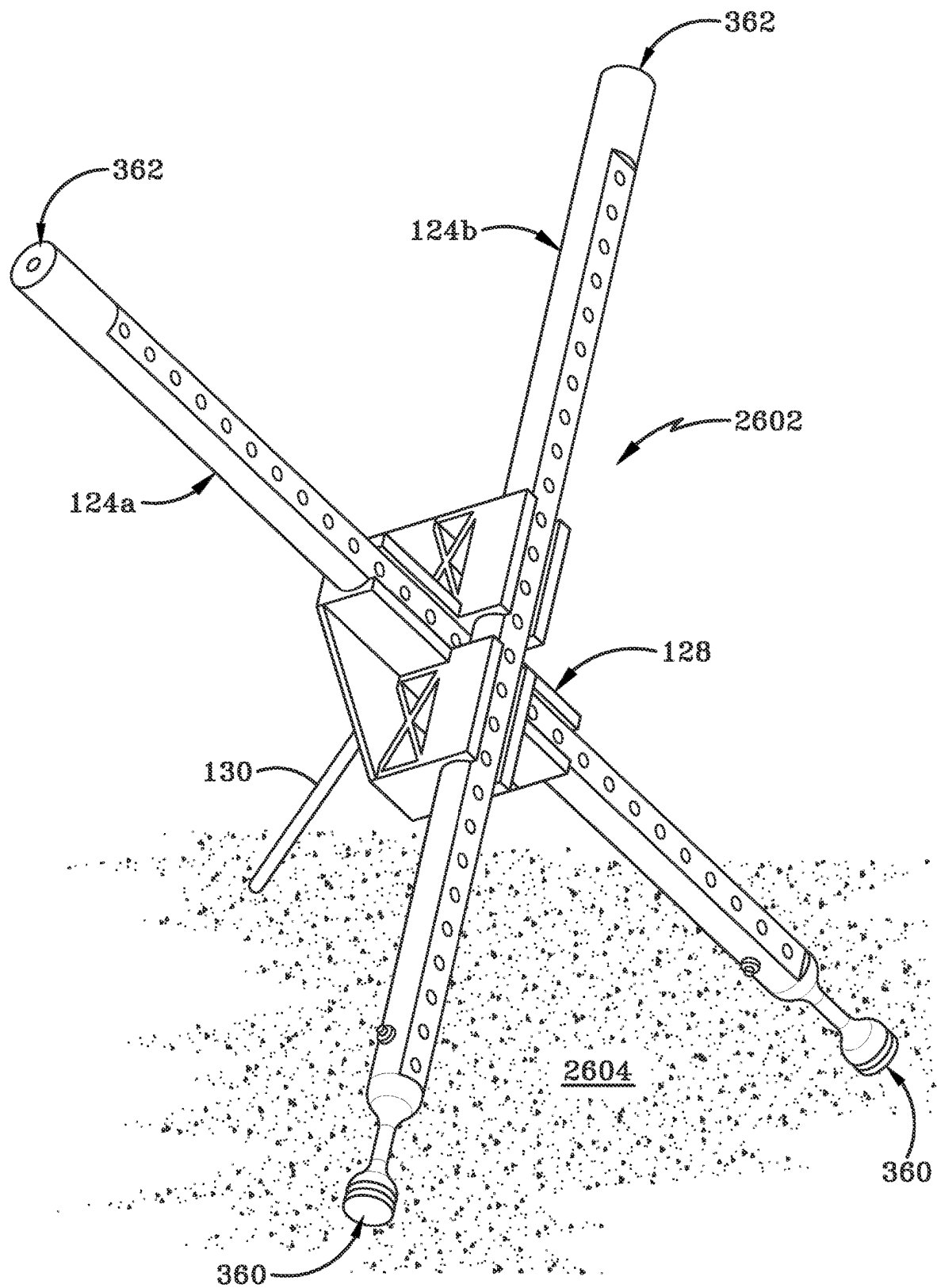


FIG. 26

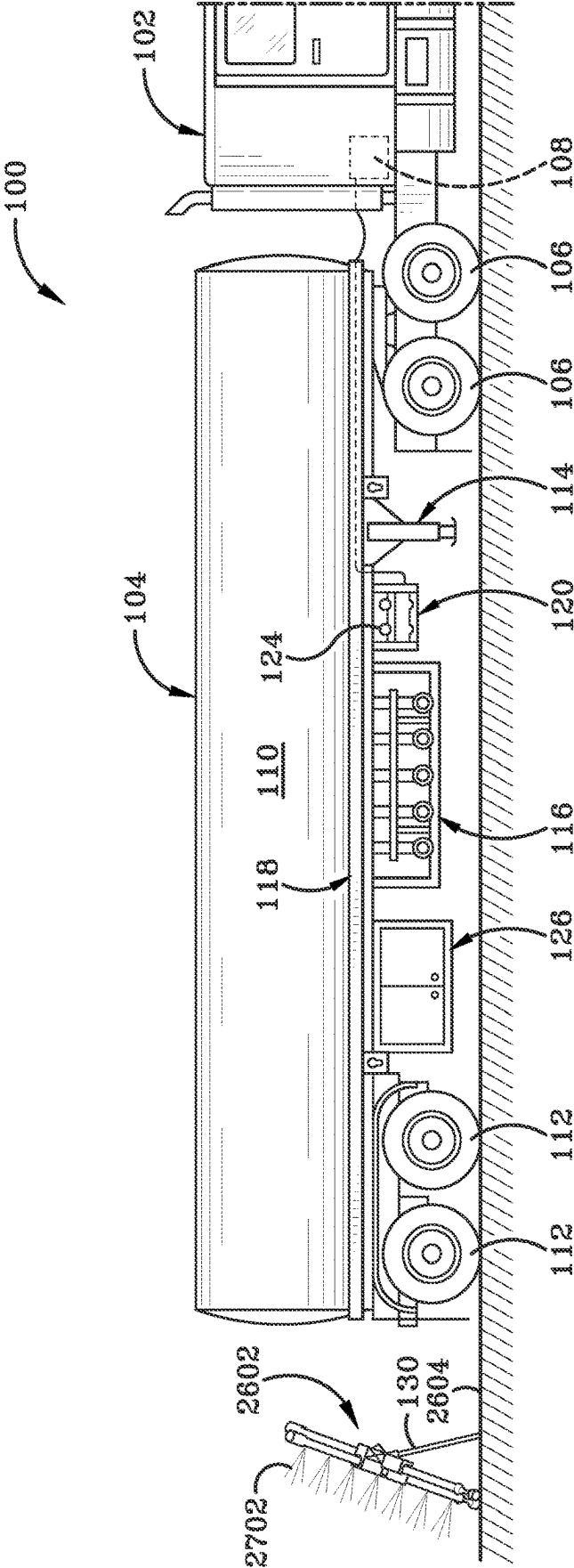


FIG. 27

1

SAFETY ZONE WITH BREAKAWAY FOR USE ON A TANKER TRUCK

CROSS REFERENCE TO RELATED APPLICATIONS

This application claims benefit of U.S. Provisional Application Ser. No. 63/013,255, filed on Apr. 21, 2020, the entire disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates generally to vehicles. More specifically, the present disclosure relates to safety systems for use with a vehicle. More specifically, the present disclosure relates to deployable safety arms for use with a vehicle.

BACKGROUND

Cargo carrying vehicles, including tanker trucks, must stop at various destinations in order to unload their cargo. Typically, after arriving at a destination, a driver must exit the cab of the truck in order to begin the unloading process. During the unloading process, the driver typically stands next to the truck. Often, the unloading process is at a high traffic location, like a gas station. As a result, when the driver is at risk of being struck by another vehicle while unloading cargo. Furthermore, the unloading process may be performed at night when the driver may be less visible to other drivers. The unloading process may require the driver's full attention, making the driver unaware of potential hazards.

SUMMARY

Currently, truck drivers must leave their vehicle to perform tasks that may divert their attention away from their surroundings. These tasks may be performed in high traffic areas putting the driver at risk from being struck by another vehicle. Thus, there is a continuous need for safety systems that protect drivers outside of their vehicle. In order to address this continuous need, the present disclosure provides safety arms for protecting drivers outside of their vehicle. As such, aspects of the present disclosure relate to safety arms for use with a vehicle.

In one aspect, an exemplary embodiment of the present disclosure may further provide a vehicle with a first safety arm and a first coupler on the first safety arm. The first coupler may couple the first safety arm to the vehicle. The first safety arm may be adapted to fall away from the first vehicle without damaging the vehicle when the vehicle contacts an object. This embodiment or another exemplary embodiment may further provide a trailer; and a cab connected to the trailer, wherein the first safety arm is coupled to the trailer. This embodiment or another exemplary embodiment may further provide a trailer; and a cab connected to the trailer, wherein the first safety arm is coupled to the cab. This embodiment or another exemplary embodiment may further provide a second safety arm; a second coupler on the second safety arm, wherein the second coupler couples the second safety arm to the vehicle and wherein the second safety arm is adapted to fall away from the vehicle without damaging the vehicle when the second safety arm contacts the object. This embodiment or another exemplary embodiment may further provide a trailer; and a cab connected to the trailer, wherein the first safety arm is coupled to the trailer and the second safety arm is coupled to the cab.

2

This embodiment or another exemplary embodiment may further provide a trailer; and a cab connected to the trailer, wherein the first safety arm and the second safety arm are coupled to the trailer. This embodiment or another exemplary embodiment may further provide a first side of the trailer; a second side of the trailer that is opposite the first side; and a nozzle assembly on the first side, wherein the first safety arm and the second safety arm are coupled to the first side on either side of the nozzle assembly. This embodiment or another exemplary embodiment may further provide a hose holder connected to the first side of the trailer adjacent to the nozzle assembly, wherein the first safety arm and the second safety arm are coupled to the hose holder. This embodiment or another exemplary embodiment may further provide a ferromagnetic surface, wherein the first coupler includes a magnet that couples the first safety arm to the ferromagnetic surface. This embodiment or another exemplary embodiment may further provide a receiver connected to the vehicle and coupled to the first coupler. This embodiment or another exemplary embodiment may further provide a cab; a trailer connected to the cab; a nozzle assembly on the trailer; and a hose holder connected to the trailer adjacent to the nozzle assembly, wherein the receiver is connected to the hose holder.

This embodiment or another exemplary embodiment may further provide wherein the first coupler is adapted to break without damaging the vehicle. This embodiment or another exemplary embodiment may further provide wherein the receiver includes a keyhole and the first coupler includes a first portion within the keyhole and a second portion outside of the keyhole, that is adapted to break when the first safety arm contacts the object. This embodiment or another exemplary embodiment may further provide wherein the first coupler is adapted to fall away from the receiver with the first safety arm without damaging the vehicle. This embodiment or another exemplary embodiment may further provide wherein the receiver includes a triangular arm receiver and the first coupler rests within the triangular arm receiver. This embodiment or another exemplary embodiment may further provide wherein the receiver includes a hemispherical surface and the first coupler includes a hemispherical member that abuts the hemispherical surface of the receiver. This embodiment or another exemplary embodiment may further provide wherein the first safety arm includes a light source.

This embodiment or another exemplary embodiment may further provide wherein the first safety arm includes a battery in electrical communication with the light source. This embodiment or another exemplary embodiment may further provide a power source; and a power outlet in electrical communication with the power source and connected to a plug of the first safety arm, wherein the plug is in electrical communication with the light source. This embodiment or another exemplary embodiment may further provide wherein the first safety arm is coupled to the vehicle without any support structure depending from the first safety arm. In another aspect, an exemplary embodiment of the present disclosure may further provide a method for detaching a safety arm from a vehicle. The method may include attaching a safety arm to a vehicle. The method may further include applying a force to the safety arm. The method may further include detaching the safety arm from the vehicle with the applied force without damaging the vehicle. This embodiment or another exemplary embodiment may further provide illuminating an area adjacent to the vehicle with a light source on the arm while the arm is attached to the vehicle. This embodiment or another exemplary embodiment may further provide moving the vehicle; contacting an

3

object with the arm while the vehicle is moving, wherein the contact applies the force to the arm; and detaching the arm from the vehicle while the vehicle is moving without damaging the vehicle. This embodiment or another exemplary embodiment may further provide breaking a coupler of the arm with the applied force, wherein breaking the coupler detaches the arm from the vehicle without damaging the vehicle. This embodiment or another exemplary embodiment may further provide wherein the arm detaches from the vehicle by falling away from the vehicle without damaging the vehicle.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

A sample embodiment of the disclosure is set forth in the following description, is shown in the drawings and is particularly and distinctly pointed out and set forth in the appended claims. The accompanying drawings, which are fully incorporated herein and constitute a part of the specification, illustrate various examples, methods, and other example embodiments of various aspects of the disclosure. It will be appreciated that the illustrated element boundaries (e.g., boxes, groups of boxes, or other shapes) in the figures represent one example of the boundaries. One of ordinary skill in the art will appreciate that in some examples one element may be designed as multiple elements or that multiple elements may be designed as one element. In some examples, an element shown as an internal component of another element may be implemented as an external component and vice versa. Furthermore, elements may not be drawn to scale.

FIG. 1 (FIG. 1) is a side view of a vehicle with multiple safety arms in a stored position.

FIG. 2 (FIG. 2) is a side view of a trailer of the vehicle depicted in FIG. 1.

FIG. 3 (FIG. 3) is a cross sectional view of the trailer depicted in FIG. 2 along the line 3-3.

FIG. 4 (FIG. 4) is a cross sectional view of a safety arm holder depicted in FIG. 3 along the line 4-4.

FIG. 5 (FIG. 5) is a side view of the trailer depicted in FIG. 2 with two safety arms removed from the trailer.

FIG. 6 (FIG. 6) is a side view of the trailer depicted in FIG. 2 with one safety arm coupled to a safety arm receiver.

FIG. 7A (FIG. 7A) is an enlarged view of a safety arm and a safety arm receiver depicted in FIG. 6.

FIG. 7B (FIG. 7B) is an enlarged view of the safety arm coupled to the safety arm receiver depicted in FIG. 6.

FIG. 7C (FIG. 7C) is cross sectional view of the safety arm coupled to the safety arm receiver depicted in FIG. 7B along the line 7C-7C.

FIG. 8 (FIG. 8) is an enlarged view of an embodiment of safety arm with a power plug coupled to a power outlet.

FIG. 9 (FIG. 9) is a side view of the trailer depicted in FIG. 2 with two safety arms coupled to two safety arm receivers.

FIG. 10 (FIG. 10) is an enlarged view of a safety arm depicted in FIG. 9 contacting an object and breaking away from a safety arm receiver.

FIG. 11 (FIG. 11) is an enlarged view of a safety arm that is magnetized to a safety arm receiver.

FIG. 12 (FIG. 12) depicts the safety arm depicted in FIG. 11 contacting an object and falling away from the safety arm receiver.

FIG. 13 (FIG. 13) is an enlarged view of a safety arm in the operational position that is magnetized to the vehicle.

4

FIG. 14 (FIG. 14) is an enlarged view of an embodiment of a safety arm a triangularly shaped coupler and a safety arm receiver.

FIG. 15 (FIG. 15) is an enlarged view of the safety arm depicted in FIG. 14 in coupled to a safety arm receiver.

FIG. 16 (FIG. 16) is a top view of the safety arm depicted in FIG. 15 along the line 16-16 coupled to a safety arm receiver.

FIG. 17 (FIG. 17) is a top view of the safety arm depicted in FIG. 15 along the line 16-16 contacting an object.

FIG. 18 (FIG. 18) is a side view of an embodiment of a safety arm with a ball lock and a safety arm receiver.

FIG. 19 (FIG. 19) is a side view of the safety arm depicted in FIG. 18 in the coupled to a safety arm receiver.

FIG. 20 (FIG. 20) is a top view of the safety arm coupled to a safety arm receiver as depicted in FIG. 19 along the line 20-20.

FIG. 21 (FIG. 21) is a top view of the safety arm and safety arm receiver depicted in FIG. 19 along the line 20-20 contacting an object.

FIG. 22 (FIG. 22) is a side view of a trailer with members of a safety device removed from the trailer.

FIG. 23 (FIG. 23) is a side view of a bracket of a safety device.

FIG. 24 (FIG. 24) is a side view of the bracket depicted in FIG. 23 that has been inverted.

FIG. 25A (FIG. 25A) is a side view of the bracket depicted in FIG. 23 coupled to a safety arm.

FIG. 25B (FIG. 25B) is a side view of the bracket and safety arm depicted in FIG. 25A wherein the safety arm has been rotated.

FIG. 25C (FIG. 25C) is a side view of the bracket depicted in FIG. 23 coupled to two safety arms.

FIG. 26 (FIG. 26) depicts a safety device formed by the bracket and safety arms depicted in FIG. 25C standing vertically on a surface.

FIG. 27 (FIG. 27) is a side view of the safety device depicted in FIG. 26 adjacent to a vehicle.

Similar numbers refer to similar parts throughout the drawings.

DETAILED DESCRIPTION

FIG. 1 depicts a vehicle 100. While the vehicle 100 is depicted as a truck, it is envisioned the vehicle 100 may be any vehicle. The vehicle 100 may include a tractor unit or a cab 102 and a trailer 104. The cab 102 may be connected to the trailer 104 to provide a pulling force for moving the vehicle 100.

The cab 102 includes one or more pairs of wheel assemblies 106 and a power source 108. The one or more wheel assemblies 106 may be adapted to support the cab 102 and the trailer 104. The power source 108 may be carried by the trailer 104.

The trailer 104 includes a generally cylindrical tank 110, one or more pairs of wheel assemblies 112, a landing gear 114, and a nozzle assembly 116. The one or more wheel assemblies 112 may be adapted to support the tank 110. Furthermore, when the vehicle 100 is stationary, the landing gear 114 may be adapted to provide further support to the tank 110. The tank 110 includes a hollow interior for transporting material. The shape and size of the tank 110 will be dependent upon the material being transported. The nozzle assembly 116 may be on a side of the tank 110 and may be in open communication with the hollow interior of the tank 110. Hoses may be selectively engaged with the nozzle assembly 116 to transfer the material (i.e., fuel,

5

liquid, grain, etc.) from the hollow interior of the tank 110 to a storage container. Although the trailer 104 is referred to herein as including a cylindrical tank 110, it is envisioned that the trailer 104 may include non-cylindrical trailers (i.e., rectangular cargo trailers). The trailer 104 further includes a first hose holder 118 connected to the tank 110. As shown in FIG. 2, the first hose holder 118 is adapted to hold one or more hoses 202. As further shown in FIG. 2, the trailer 104 includes a second hose holder 204 connected to the tank 110 and adapted to hold one or more hoses 202.

Returning to FIG. 1, the trailer 104 further includes and carries a safety arm holder 120 and one or more safety arm receivers. As will be discussed in further detail below, in one embodiment, the safety arm holder 120 is connected to the first hose holder 118 and the second hose holder 204. In another embodiment, the safety arm holder 120 may be mounted anywhere on any outer surface of the vehicle 100 (i.e., adjacent to the nozzle assembly 116). The safety arm holder 120 may be adapted to retain one or more safety arms 124 outside of the vehicle 100 while the vehicle 100 is stationary or moving. While the safety arm holder 120 is shown holding four safety arms 124, it is envisioned that the safety arm holder 120 may hold any number of safety arms 124 (i.e., 1, 3, 6, 9, 10, etc.). In one embodiment (FIG. 1), the safety arm holder 120 may be in electrical communication with the power source 108. Although the power source 108 is shown within cab 102, it is envisioned that the power source 108 may be located elsewhere on the vehicle 100. It is further envisioned that the power source 108 may be external from the vehicle 100.

As will be discussed in further detail below, the safety arm receivers are adapted to couple to a safety arm 124 such that, when coupled, a safety arm receiver provides vertical and horizontal support for a safety arm 124 so that a safety arm 124 may extend from the vehicle 100 without any support structure depending from the safety arm 124.

The trailer 104 further includes a bracket holder 126. Although the bracket holder 126 is shown connected to the first hose holder 118, it is envisioned that the bracket holder 126 may be connected to the second hose holder 204 or the tank 110. Furthermore, the bracket holder 126 may be adapted to contain a bracket 128 and a bracket support 130. As will be discussed in further detail below, the bracket 128 may be adapted to receive one or more safety arms 124 and the bracket support 130. The bracket support 130 may be adapted to support the bracket 128.

FIG. 2 depicts the tank 110. The tank 110 includes a first end 206, a second end 208, a first side 210, a second side 212, a top side 214, and a bottom side 216. The first end 206 faces the cab 102 and the second end 208 is opposite the first end 206. A central longitudinal axis 218 extends between and through the first end 206 and the second end 208 thereby defining a longitudinal direction between the first end 206 and the second end 208. The first side 210 is opposite the second side 212 and a central transverse axis 220 extends between and through the first side 210 and the second side 212 thereby defining a transverse direction between the first side 210 and the second side 212. The top side 214 is opposite the bottom side 216 and a central vertical axis 222 extends between and through the top side 214 and the bottom side 216 thereby defining a vertical direction between the top side 214 and the bottom side 216. The central longitudinal axis 218, the central transverse axis 220, and the central vertical axis 222 are oriented at right angles relative to one another.

FIG. 3 depicts safety arms 124 in a stored position. The safety arm holder 120 slidably receives and supports the

6

safety arms 124 and the safety arms 124 extend parallel to the central transverse axis 220 and perpendicular to the central longitudinal axis 218.

The safety arm holder 120 includes a first holder 302 a distance away from a second holder 304. The distance between the first holder 302 and the second holder 304 is less than a length of a safety arm 124. As such, a safety arm 124 may be carried and retained by the first holder 302 and the second holder 304 when the vehicle 100 is stationary or moving.

In one embodiment, the first holder 302 may be connected to the first side 210 of the tank 110 and the second holder 304 may be connected to the second side 212 of the tank 110. In this embodiment, the first holder 302 may be connected to a bottom surface 306 of the first hose holder 118 and the second holder 304 may be connected to a bottom surface 308 of the second hose holder 204. The first holder 302 and the second holder 304 are further connected to a mounting member 310 and 312 respectively. The mounting members 310 and 312 are connected to the tank 110. In another embodiment, the first holder 302 and the second holder 304 may be anywhere on the vehicle 100 (i.e., on the first side 210 of the tank 110, on the second side 212 of the tank 110, etc.).

The first holder 302 includes a first end 314, a second end 316 opposite the first end 314, a top end 318, and a bottom end 320 opposite the top end 318. The first end 314 extends longitudinally relative to the central vertical axis 222 from the bottom surface 306 to the bottom end 320 and the second end 316 extends longitudinally relative to the central vertical axis 222 from the mounting member 310 to the bottom end 320. Opposing inner surfaces 322 extend longitudinally relative to the central vertical axis 222 from the top end 318 to the bottom end 320 and longitudinally relative to the central longitudinal axis 218 from the first end 314 to the second end 316. The first end 314 and the second end 316 are open ended such that when a safety arm 124 is moved in either direction of arrow A, a safety arm 124 may pass through the first holder 302.

The first holder 302 further includes a first rack 324 and a second rack 326. The first rack 324 is vertically above second rack 326. Stated another way, the second rack 326 is vertically below the first rack 324. The first rack 324 and the second rack 326 extend between the first end 314 and the second end 316 of the first holder 302. Furthermore, the first rack 324 and the second rack 326 are adapted to slidably receive and support one or more safety arms 124.

The second holder 304 includes a first end 328, a second end 330 opposite the first end 328, a top end 332, and a bottom end 334 opposite the top end 332. The first end 328 extends longitudinally relative to the central vertical axis 222 from the bottom surface 308 to the bottom end 334 and the second end 330 extends longitudinally relative to the central vertical axis 222 from the mounting member 312 to the bottom end 334. Opposing inner surfaces 336 extend longitudinally relative to the central vertical axis 222 from the top end 332 to the bottom end 334 and longitudinally relative to the central vertical axis 222 from the first end 328 to the second end 330.

The second holder 304 further includes a first rack 338 and a second rack 340. The first rack 338 is vertically above second rack 340. Stated another way, the second rack 340 is vertically below the first rack 338. The first rack 338 and the second rack 340 extend between the first end 328 and the second end 330 of the second holder 304. The first rack 338 and the second rack 340 are adapted to slidably receive and support one or more safety arms 124.

In one embodiment only the second end 330 of the second holder 304 is open ended. In this embodiment, when placing a safety arm 124 into the stored position, a safety arm 124 enters the safety arm holder 120 from the first side 210 of the tank 110, passes through the first end 314 of the first holder 302 and through the second end 330 of the second holder 304. Stated another way, a safety arm 124 first enters the first holder 302 then enters the second holder 304.

In another embodiment, the first end 328 and the second end 330 of the second holder 304 are open ended. In this embodiment, when placing a safety arm 124 in a stored position, a safety arm 124 may enter the safety arm holder 120 from the first side 210 or the second side 212 of the tank 110. Stated another way, a safety arm 124 may first pass through the first holder 302 and then into the second holder 304 or may first pass through the second holder 304 and then into the first holder 302.

In one embodiment (FIG. 3), the second holder 304 may be in electrical communication with the power source 108. In this embodiment, the second holder 304 includes a charging dock 342 and a safety arm 124 includes and carries a battery 344. The charging dock 342 is carried by the trailer 104 and is in electrical communication with the power source 108 and is adapted to charge the battery 344 of a safety arm 124 when a safety arm 124 is connected to the charging dock 342. In this embodiment, the battery 344 is in electrical communication and supplies power to a light source 345 of a safety arm 124.

The charging dock 342 includes a number of receptacles or channels 346. The number of channels 346 may depend upon the number of safety arms 124 that the safety arm holder 120 is adapted to retain. For example, in the embodiment shown in FIG. 2, the safety arm holder 120 is adapted to retain four safety arms 124. Hence, the charging dock 342 may contain four channels 346. It is envisioned that the charging dock 342 may contain any number of channels 346 (i.e., 1, 3, 6, 9, 10, etc.). The channels 346 are defined by a surface 348. The surface 348 extends from a side surface 352 to an opening 354. The surface 348 defines a shape of a channel 346. The shape of a channel 346 may correspond to a shape and a dimension of a safety arm 124. For example, in the embodiment shown in FIG. 4, a safety arm 124 has a circular shape. As such, a channel 346 may be circular with a diameter that corresponds to a diameter of a safety arm 124. Due to the corresponding shape and dimension, when a safety arm 124 is in the stored position, a portion of the safety arm 124 nests within the channel 346.

In the embodiment shown in FIG. 3, a channel 346 further includes a charging connector 358. The charging connector 358 is in electrical communication with the power source 108. Furthermore, a safety arm 124 includes a first end 360, a second end 362 opposite the first end 360 defining a longitudinal direction there between. In the embodiment shown in FIG. 3, a charging receptacle 364 extends inward from the second end 362 towards the first end 360 and is in electrical communication with the battery 344. The charging receptacle 364 has a complementary shape to the charging connector 358 and is adapted to slidably receive the charging connector 358 thereby connecting the charging connector 358 to the charging receptacle 364. When a safety arm 124 is in the stored position, the second end 362 of a safety arm 124 abuts the side surface 352 of the channel 346 and the charging receptacle 364 connects to the charging connector 358. The complementary shapes of the charging connector 358 and the charging receptacle 364 provides a frictional interference fit between the charging connector 358 and the charging receptacle 364. When the charging connector 358

is connected to the charging receptacle 364, the battery 344 is in electrical communication with the power source 108 which allows the power source 108 to charge the battery 344. The charging dock 342 may charge any number of safety arms 124 simultaneously.

As will be discussed in further detail below, a safety arm 124 further includes coupler that is adapted to attach or couple a safety arm 124 to the vehicle 100. A safety arm 124 is adapted to fall away from the vehicle 100 without damaging the vehicle 100 (i.e., the cab 102 or the trailer 104).

FIG. 4 depicts the second holder 304. As shown in FIG. 4, the first rack 338 and the second rack 340 of the second holder 304 are structurally identical. For the sake of brevity, reference numerals on the first rack 338 correspond to identical elements on the second rack 340. Furthermore, it is understood that the first rack 338 and the second rack 340 are structurally identical to the first rack 324 and the second rack 326 of the first holder 302. The first rack 338 includes a first surface 402 and a second surface 404 opposite the first surface 402. The first surface 402 includes a number of grooves 406. The number of grooves 406 will be dependent upon a number of safety arms 124 that the safety arm holder 120 is adapted to retain. For example, in the embodiment shown in FIG. 2, the safety arm holder 120 is adapted to retain four safety arms 124. Hence, the first rack 338 contains two grooves 406 and the second rack 340 contains two grooves 406 for a total of four grooves 406. It is envisioned that either the first rack 338 or the second rack 340 may contain any number of grooves 406 (i.e., 1, 3, 6, 9, 10, etc.). It is further envisioned that the first holder 302 and the second holder 304 may contain any number of racks (1, 3, 6, 9, 10, etc.).

Furthermore, a shape and dimension of a groove 406 corresponds to a shape and dimension of a safety arm 124. For example, in the embodiment shown in FIG. 4, a groove 406 is semi-circular. The semi-circular shape corresponds to the circular shape and dimension of the safety arm 124. Due to the corresponding shape and dimension, when a safety arm 124 is in the stored position, the safety arm 124 nests within the groove 406. When the second holder 304 includes a charging dock 342, the surface 348 of a channel 346 is flush with groove 406 which allows a safety arm 124 to slide into the channel 346 from a first rack 338 or a second rack 340.

FIG. 5 depicts safety arms 124 removed from the trailer 104. In order to remove safety arms 124 from the trailer 104, safety arms 124 may be slidably moved in the direction of arrow B.

FIG. 6 depicts a breakaway assembly 600. In this embodiment the breakaway assembly 600 includes a coupler 602 at the first end 360 of a safety arm 124 and a safety arm receiver 604 on the vehicle 100.

In this embodiment, the coupler 602 couples to safety arm receiver 604 by moving the safety arm 124 in the direction of arrow C. The safety arm receiver 604 includes a keyhole 606 with a first portion 608 and a second portion 610. The keyhole 606 receives the coupler 602 of a safety arm 124.

FIGS. 7A-7C depict a safety arm 124 with coupler 602 and the safety arm receiver 604. The safety arm receiver 604 includes a first wall 605 and a second wall 607 that extend from the bottom surface 306 of the first hose holder 118. The second wall 607 includes an outer surface 612 and an inner surface 614. The keyhole 606 extends from the outer surface 612 to the inner surface 614. Although the safety arm receiver 604 is shown connected to the first hose holder 118, it is envisioned that the safety arm receiver 604, and other

embodiments of arm receivers, may be connected to the second hose holder 204, the tank 110, or the cab 102.

The coupler 602 includes a first circular wall 620. The first circular wall 620 includes a first circular surface 622 and a second circular surface 624 opposite the first circular surface 622. The coupler 602 further includes a second circular wall 626 that extends from the second circular surface 624 to a circular surface 628 of a first tapered portion 630 of the coupler 602. The first circular wall 620 has a greater diameter than the second circular wall 626. The first tapered portion 630 extends from the circular surface 628 to a break portion 632 of the coupler 602. The break portion 632 extends from the first tapered portion 630 to a second tapered portion 634.

FIG. 7B depicts a safety arm 124 coupled to safety arm receiver 604. The keyhole 606 is dimensioned so that the first circular wall 620 may pass through the first portion 608 of the keyhole 606. The first circular wall 620 may pass through the first portion 608 by moving a safety arm 124 in the direction of arrow D. When the first circular wall 620 passes through the first portion 608 and the second circular wall 626 aligns with the second portion 610 of the keyhole 606, the safety arm 124 is then moved in the direction of arrow E thereby coupling the coupler 602 to the safety arm receiver 604.

FIG. 7C further depicts a safety arm 124 coupled to a safety arm receiver 604. The keyhole 606 extends from the outer surface 612 of the first wall 605 of the safety arm receiver 604 to the inner surface 614 of the first wall 605. When the coupler 602 is coupled to the safety arm receiver 604, the second circular surface 624 abuts the inner surface 614 of the first wall 605, the circular surface 628 abuts the outer surface 612 of the first wall 605 and the second circular wall 626 nests within the second portion 610 of the keyhole 606.

FIG. 8 depicts an embodiment wherein a power outlet 802 is connected to the first hose holder 118 and adjacent to the safety arm receiver 604. In another embodiment, the power outlet 802 may be connected to the second hose holder 204 or anywhere else on the vehicle 100. The power outlet 802 is in electrical communication with the power source 108. In this embodiment, the power outlet 802 is adapted to receive a plug 804 of a safety arm 124. The plug 804 is connected to a wire 806 that places the plug 804 in electrical communication with the light source 345. As such, when the power outlet 802 receives the plug 804, the power source 108 supplies power to the light source 345 of the safety arm 124.

FIG. 9 depicts two safety arms 124 in an operational position. When in the operational position, one safety arm 124 or two safety arms 124, as depicted in FIG. 9, extend outwardly from the vehicle 100 thereby creating a safety zone. While FIG. 9 depicts two safety arms 124 coupled to the trailer 104, it is understood that any number of safety arms 124 may be coupled to the vehicle 100 (i.e., the trailer 104 or the cab 102). The safety arms 124 include a first side 902, a second side 904 opposite the first side 902, and power switch 906. While FIG. 9 depicts the light source 345 on the first side 902 and the second side 904 of the safety arm 124, it is envisioned that the safety arm 124 may include the light source 345 on only the first side 902 or only the second side 904. In this embodiment, the two safety arms 124 are coupled to the first side 210 and adjacent to the nozzle assembly 116.

The power switch 906 controls the flow of electricity from a battery 344 or the power source 108 to the light source 345. A user may engage the power switch 906 a number of times to power a light source 345. For example, when a safety arm

124 includes the light source 345 on the first side 902 and the second side 904 and is off (i.e., no light is being emitted from the light source 345 on either the first side 902 or the second side 904), a user may engage the power switch 906 a first time to cause a light source 345 to emit light from the first side 902 and the second side 904. After engaging the power switch 906 a first time to emit light from both the first side 902 and the second side 904, the user may engage the power switch 906 a second time to emit light from only one side (i.e., the first side 902 or the second side 904) of the safety arm 124. The user may then engage the power switch 906 a third time to power off the safety arm 124 (i.e., no light is being emitted from the light source 345 on either the first side 902 or the second side 904). When on, the light source 345 emits light 908 that illuminates an area adjacent to the vehicle 100. The light source 345 may be any suitable source of illumination including, but not limited to, at least one strip of light emitting diodes.

FIG. 10 depicts a safety arm 124 contacting an object 1002 (i.e., a stationary pole). In one embodiment, the coupler 602 may be formed from a suitably rigid material so as to support the safety arm 124 when the safety arm 124 is coupled to the safety arm receiver 604 without any support structure depending from the safety arm 124. Furthermore, the integral structure of the coupler 602 may be formed from multiple elements having similar configurations as one having ordinary skill in the art would understand.

In one embodiment, the first tapered portion 630, the break portion 632, and the second tapered portion 634 are formed of the same material. In another embodiment, the first tapered portion 630, the break portion 632, and the second tapered portion 634 are formed of different materials. In this embodiment, the break portion 632 is formed of a material that is less rigid, and therefore more susceptible to breaking, than a material that forms a remainder of the coupler 602 including the first tapered portion 630 and the second tapered portion 634. Furthermore, the break portion 632 has a smaller diameter than the first tapered portion 630 and the second tapered portion 634. As such, the break portion 632 is less rigid and more susceptible to breaking than the remainder of the coupler 602 including first tapered portion 630 or the second tapered portion 634.

In one example, when the vehicle 100 moves in the direction of arrow F and the object 1002 contacts the safety arm 124, a force is applied to the safety arm 124 causing the safety arm 124 to move in the direction of arrow G. This force may cause the coupler 602 to break within the break portion 632. Due to the location of the break portion 632, when the break portion 632 breaks, the vehicle 100 is undamaged. When the break portion 632 breaks, the safety arm 124 falls away from the vehicle 100.

FIG. 11 depicts a breakaway assembly 1100. In this embodiment, the breakaway assembly 1100 includes a coupler 1102 at the first end 360 of a safety arm 124 and a safety arm receiver 1104 on the vehicle 100.

In this embodiment, the coupler 1102 includes a magnet that couples to a safety arm receiver 1104 that is formed from a suitable ferromagnetic material. The safety arm receiver 1104 includes a first wall 1106 that includes a surface 1108. The coupler 1102 includes a surface 1110 that defines the first end 360 of a safety arm 124 and magnetizes the coupler 1102 to the surface 1108 of the safety arm receiver 1104. In this embodiment, the coupler 1102 may be suitably strong to support the safety arm 124 when the safety arm 124 is coupled to the safety arm receiver 1104 without any support structure depending from the safety arm 124.

11

FIG. 12 depicts a safety arm 124 contacting the object 1002. In this embodiment, the coupler 1102 may not be strong enough to support the safety arm 124 when the safety arm 124 contacts the object 1002. For example, when the vehicle 100 moves in the direction of arrow H and the object 1002 contacts the safety arm 124, a force is applied to the safety arm 124 causing the safety arm 124 to move in the direction of arrow I. This force causes the coupler 1102 to demagnetize from the safety arm receiver 1104. When the coupler 1102 demagnetizes from the safety arm receiver 1104 due to a force applied to the safety arm 124, the safety arm 124 falls away from the vehicle 100 with the coupler 1102 and without damaging the vehicle 100. In this embodiment, the safety arm 124 may be undamaged and the safety arm 124 may be recoupled to the safety arm receiver 1104.

Although FIGS. 11 and 12 depict the coupler 1102 coupled to a safety arm receiver 1104, it is envisioned that the coupler 1102 may couple a safety arm 124 to any ferromagnetic surface of the vehicle 100. For example, FIG. 13 depicts a breakaway assembly 1300. In this embodiment, the breakaway assembly 1300 includes the coupler 1102 and a safety arm receiver 1304. In this embodiment, the safety arm receiver 1304 may be a disk extending from a surface 118A of the first hose holder 118. While the safety arm receiver 1304 is depicted as extending from the first hose holder 118, it is envisioned that the safety arm receiver 1304 may extend any surface of the vehicle 100. The safety arm receiver 1304 may be made formed of a ferromagnetic material or a magnet that attracts the coupler 1102.

FIG. 14 depicts a breakaway assembly 1400. In this embodiment, the breakaway assembly 1400 includes a coupler 1402 and a safety arm receiver 1404. The coupler 1402 is generally triangular in shape and the safety arm receiver 1404 has a complimentary triangular shape.

In this embodiment, the coupler 1402 defines the first end 360 of a safety arm 124. A neck 1406 extends from the safety arm 124 to a first surface 1408 of the coupler 1402. The first surface 1408 extends between a second surface 1410 and a third surface 1412. A fourth surface 1414 and a fifth surface 1416 opposite the fourth surface 1414 extend from and generally perpendicular to the first surface 1408 to a first angled surface 1418 and a second angled surface 1420. The fourth surface 1414 and the fifth surface 1416 extend between the second surface 1410 and the third surface 1412. The first angled surface 1418 is opposite the second angled surface 1420. The first angled surface 1418 and the second angled surface 1420 extend from the fourth surface 1414 and the fifth surface 1416 respectively to a rounded corner 1422. Furthermore, the first angled surface 1418, the second angled surface 1420, and the rounded corner 1422 extend between the second surface 1410 the third surface 1412.

The safety arm receiver 1404 is adapted to receive the coupler 1402 and includes a bottom wall 1405 and a top wall 1407. The safety arm receiver 1404 further includes a first outer edge 1423, a second outer edge 1424 opposite the first outer edge 1423, a third outer edge 1426, and a fourth outer edge 1428 opposite the third outer edge 1426. The third outer edge 1426 and the fourth outer edge 1428 extend between the bottom wall 1405 and the top wall 1407. The first outer edge 1423 and the second outer edge 1424 extend between and generally perpendicular to the third outer edge 1426 and the fourth outer edge 1428. The third outer edge 1426 and the fourth outer edge 1428 extend between and generally perpendicular to the first outer edge 1423 and the second outer edge 1424. The first outer edge 1423, the

12

second outer edge 1424, the third outer edge 1426, and the fourth outer edge 1428 define an aperture 1411 that receives the coupler 1402.

The safety arm receiver 1404 further includes an angled wall 1409. The angled wall 1409 includes a first angled surface 1430, a second angled surface 1432 opposite the first angled surface 1430, a rounded corner 1434. The safety arm receiver 1404 further includes a first surface 1436, and a second surface 1438. The first angled surface 1430 and the second angled surface 1432 extend from the third outer edge 1426 and the fourth outer edge 1428 respectively to the rounded corner 1434. The first angled surface 1430, the second angled surface 1432, and the rounded corner 1434 extend between the first surface 1436 and the second surface 1438. The first surface 1436 and the second surface 1438 extend between the first angled surface 1430 and the second angled surface 1432. Furthermore, the first surface 1436 extends between the first outer edge 1423 and the rounded corner 1434 and the second surface 1438 extends between the second outer edge 1424 and the rounded corner 1434.

As depicted in FIGS. 15 and 16, when the coupler 1402 is coupled to the safety arm receiver 1404, the coupler 1402 nests within the safety arm receiver 1404. When the coupler 1402 nests within the safety arm receiver 1404, the second surface 1410 of the coupler 1402 abuts the first surface 1436 of the safety arm receiver 1404, the third surface 1412 of the coupler 1402 abuts the second surface 1438 of the safety arm receiver 1404, the rounded corner 1422 of the coupler 1402 abuts the rounded corner 1434 of the safety arm receiver 1404, the first angled surface 1418 of coupler 1402 abuts the first angled surface 1430 of the safety arm receiver 1404, and the second angled surface 1420 of coupler 1402 abuts the second angled surface 1432 of the safety arm receiver 1404.

As depicted in FIG. 16, the first angled surface 1418 is not as long as the first angled surface 1430 and the second angled surface 1420 is not as long as the second angled surface 1432. As a result, the fourth surface 1414 and the fifth surface 1416 of coupler 1402 do not abut a surface of the safety arm receiver 1404.

FIG. 17 depicts a safety arm 124 contacting the object 1002. In this embodiment, when the safety arm 124 contacts the object 1002 the safety arm 124 falls away from the vehicle 100 with coupler 1402. For example, when the vehicle 100 moves in the direction of arrow J and the object 1002 contacts the safety arm 124, a force is applied to the safety arm 124 causing the safety arm 124 to move in the direction of arrow K. This force causes coupler 1402 to detach or decouple from the safety arm receiver 1404. When coupler 1402 decouples from the safety arm receiver 1404 due to a force applied to the safety arm 124, the safety arm 124 falls away from the vehicle 100 without damaging the vehicle 100. In this embodiment, the safety arm 124 may be undamaged and the coupler 1402 may be recoupled to the safety arm receiver 1404.

FIG. 18 depicts a breakaway assembly 1800. In this embodiment, the breakaway assembly 1800 includes a coupler 1802 at the first end 360 of a safety arm 124 and a safety arm receiver 1803 on the vehicle 100. While FIG. 18 depicts the safety arm receiver 1803 on the first hose holder 118, it is envisioned that the safety arm receiver 1803 may be coupled to the second hose holder 204.

The coupler 1802 includes an extension 1805 that extends from the safety arm 124. The coupler 1802 further includes a hook 1807 that extends from the extension 1805. The hook 1807 includes a first leg 1809 and a second leg 1811 that is perpendicular to the first leg 1809. The coupler 1802 further

13

includes a first surface **1804** of the coupler **1802** extends longitudinally relative to the central vertical axis **222** from a first surface **1806** of the extension **1805** to a first rounded corner **1808**.

The first rounded corner **1808** convexly curves downward relative to the central vertical axis **222** approximately 90° from the first surface **1804** to a second surface **1810** of the coupler **1802**. The second surface **1810** extends longitudinally relative to the central transverse axis **220** from the first rounded corner **1808** to a third surface **1812** of the coupler **1802**. The third surface **1812** extends longitudinally relative to the central vertical axis **222** from the second surface **1810** to a fourth surface **1814**. The fourth surface **1814** extends longitudinally relative to the central transverse axis **220** from the third surface **1812** to a fifth surface **1816** of the coupler **1802**. The fifth surface **1816** extends longitudinally relative to the central vertical axis **222** from the fourth surface **1814** to a sixth surface **1818**. The sixth surface **1818** extends longitudinally relative to the central transverse axis **220** from the fifth surface **1816** to a seventh surface **1820**. The seventh surface **1820** extends longitudinally relative to the central vertical axis **222** from the sixth surface **1818** to an eighth surface **1822**. The eighth surface **1822** extends longitudinally relative to the central transverse axis **220** from the seventh surface **1820** to a second rounded corner **1824**. The second rounded corner **1824** concavely curves downward relative to the central vertical axis **222** from the eighth surface **1822** to a terminal surface **1826** of the safety arm **124**.

The first surface **1804**, the third surface **1812**, the fifth surface **1816**, the seventh surface **1820**, and the terminal surface **1826** extend generally parallel relative to the central vertical axis **222**. The second surface **1810**, the fourth surface **1814**, the sixth surface **1818**, and the eighth surface **1822** extend generally perpendicular relative to the central vertical axis **222**.

A hemispherical member **1828** extends longitudinally relative to the central vertical axis from the eighth surface **1822**. As will be discussed in further detail below, the hemispherical member **1828** is adapted to nest within the safety arm receiver **1803**. The third surface **1812**, the fourth surface **1814**, the fifth surface **1816**, the sixth surface **1818**, and the seventh surface **1820** define an opening **1830**. As will be discussed in further detail below, when the coupler **1802** is coupled to the safety arm receiver **1803**, a portion of the safety arm receiver **1803** nests within the opening **1830**.

The safety arm receiver **1803** includes a post member **1832** and a hemispherical receptacle **1834**. The post member **1832** includes a first leg **1831** and a second leg **1833** that is perpendicular to the first leg **1831**. The safety arm receiver **1803** further includes a first wall **1836**. The first wall **1836** extends longitudinally relative to the central vertical axis **222** from the bottom surface **306** of the first hose holder **118**. The first wall **1836** includes a surface **1840** that extends longitudinally relative to the central vertical axis **222** from the bottom surface **306** to a second wall **1842**. The first wall **1836** extends generally parallel to the central vertical axis **222** and the second wall **1842** extends generally perpendicular to the central vertical axis **222**. The first wall **1836**, the second wall **1842**, and the bottom surface **306** define an inlet **1844**. As will be discussed in further detail below, in order to couple the coupler **1802** to the safety arm receiver **1803**, the coupler **1802** passes through the inlet **1844**.

The post member **1832** extends longitudinally relative to the central vertical axis **222** from a surface **1846** of the second wall **1842**. The post member **1832** includes a first surface **1848**. The first surface **1848** extends longitudinally

14

relative to the central vertical axis **222** to a second surface **1850**. The second surface **1850** extends longitudinally relative to the central transverse axis **220** to a rounded surface **1852**. The rounded surface **1852** concavely curves upward relative to the central vertical axis **222** approximately 180° from the second surface **1850** to a third surface **1854**. The third surface **1854** extends longitudinally relative to the central transverse axis **220** to a fourth surface **1856**. The fourth surface **1856** extends longitudinally relative to the central vertical axis **222** to the surface **1846** of the second wall **1842**. The first surface **1848** and the third surface **1854** extend generally parallel to the central vertical axis **222** while the second surface **1850** and the fourth surface **1856** extend generally perpendicular to the central vertical axis **222**.

The hemispherical receptacle **1834** includes a first surface **1858** that extends longitudinally relative to the central vertical axis **222** from the surface **1846** of the second wall **1842** to a second surface **1860**. The second surface **1860** extends longitudinally relative to the central transverse axis **220** from the first surface **1858** to a third surface **1862**. Furthermore, the second surface **1860** includes a hemispherical surface **1864**. As will be discussed in further detail below, the hemispherical surface **1864** is adapted to receive the hemispherical member **1828** of the coupler **1802**. The third surface **1862** extends longitudinally relative to the central vertical axis from the second surface **1860** to the surface **1846** of the second wall **1842**. Furthermore, the first surface **1858** of the hemispherical receptacle **1834** abuts the fourth surface **1856** of the post member **1832**.

FIGS. **19** and **20** depict the coupler **1802** coupled to the safety arm receiver **1803**. When the safety arm receiver **1803** is coupled to coupler **1802**, the second surface **1850**, rounded surface **1852**, and the third surface **1854** of the safety arm receiver **1803** nest within the opening **1830** of the coupler **1802**. In this position, second surface **1850** of the safety arm receiver **1803** abuts the fourth surface **1814** of the coupler **1802** and the third surface **1854** of the safety arm receiver **1803** abuts the sixth surface **1818** of the coupler **1802**. Furthermore, when the coupler **1802** is coupled to the safety arm receiver **1803**, the hemispherical member **1828** of the coupler **1802** nests within and abuts the hemispherical surface **1864** of the safety arm receiver **1803**.

FIG. **21** depicts a safety arm **124** with the coupler **1802** contacting the object **1002**. When the vehicle **100** moves in the direction of arrow **L** and the object **1002** contacts the safety arm **124**, a force is applied to the safety arm **124** causing the safety arm **124** rotate in the direction of arrow **M**. When the safety arm **124** is in the position depicted in FIG. **21** and the vehicle **100** continues to move in the direction of arrow **L**, the safety arm **124** moves in the direction of arrow **N** and detaches or decouples from safety arm receiver **1803**. When decoupled, the safety arm **124** and the coupler **1802** fall away from the vehicle **100** without damaging the vehicle **100**. In this embodiment, the safety arm **124** may be undamaged and the safety arm **124** may be recoupled to the safety arm receiver **1803**.

FIG. **22** depicts a first safety arm **124a**, a second safety arm **124b**, the bracket **128**, and the bracket support **130** being removed from the vehicle **100**. In order to remove the first safety arm **124a** and the second safety arm **124b**, the first safety arm **124a** and the second safety arm **124b** are moved in the direction of arrow **O**. In order to remove the bracket **128** and the bracket support **130** from the bracket holder **126**, first a first door **2202** and a second door **2204** are

15

opened and then the bracket 128 and the bracket support 130 are moved in the direction of arrow P and removed from the bracket holder 126.

FIG. 23 depicts the bracket 128. The bracket 128 includes a first side 128A, a second side 128B opposite the first side 128A, a first safety arm holder 2302 and a second safety arm holder 2304. The first safety arm holder 2302 extends generally parallel to a first horizontal axis 2306 and the second safety arm holder 2304 extends generally parallel to a second horizontal axis 2308. The first horizontal axis 2306 and the second horizontal axis 2308 intersect a central vertical axis 2309 at right angles. Stated another way, the first horizontal axis 2306 and the second horizontal axis 2308 are perpendicular to the central vertical axis 2309. The first safety arm holder 2302 extends generally perpendicular to the second safety arm holder 2304.

The bracket 128 is octagonal in shape. A first surface 2310 of the first safety arm holder 2302 defines a first side 2312 of the bracket 128. A second surface 2314 opposite the first surface 2310 defines a second side 2316 of the first safety arm holder 2302. The first side 2312 of the bracket 128 is opposite the second side 2316 of the bracket 128. A semi-circular surface 2318 extends from the first surface 2310 to the second surface 2314 of the first safety arm holder 2302 and defines a channel 2320. As will be discussed in further detail below, the channel 2320 may retain a safety arm 124.

The first surface 2310 and the second surface 2314 extend longitudinally relative to the central vertical axis to a third surface 2322 and a fourth surface 2324 of the first safety arm holder 2302. The third surface 2322 extends to fifth surface 2326 of the first safety arm holder 2302 and the fourth surface 2324 extends to a sixth surface 2328 of the first safety arm holder 2302. The fifth surface 2326 and the sixth surface 2328 define an opening 2303 of the first safety arm holder 2302. The fifth surface 2326 and the sixth surface 2328 extend to the semi-circular surface 2318. The bracket 128 further includes a base wall 2311. The base wall 2311 includes a first surface 2332. A seventh surface 2330 of the first safety arm holder 2302 extends longitudinally relative to the central vertical axis 2309 from the first surface to the third surface 2322. An eighth surface 2334 extends from the first surface 2332 to the fourth surface 2324. The first surface 2310, the second surface 2314, the fifth surface 2326, the sixth surface 2328, the seventh surface 2330, and the eighth surface 2334 extend generally perpendicular to the first horizontal axis 2306. The third surface 2322 and the fourth surface 2324 extend generally perpendicular to the first horizontal axis 2306.

The second safety arm holder 2304 includes a first portion 2336, a structurally identical second portion 2338, and an opening 2307 between the first portion 2336 and the second portion 2338. The first portion 2336 includes a first surface 2340. The first surface 2340 defines a third side 2342 of the bracket 128 and the second portion 2338 includes a first surface 2343 that define a fourth side 2345 of the bracket 128.

The third side 2342 and the fourth side 2345 are approximately 90° offset from the first side 2312 and the second side 2316 relative to the central vertical axis 2309. The first portion 2336 further includes a second surface 2344 opposite the first surface 2340. A semi-circular surface 2346 extends from the first surface 2340 to the second surface 2344 of the second safety arm holder 2304 and defines a channel 2348. The channel 2348 extends from the first surface 2340 of the first portion 2336 to the first surface 2343 of the second portion 2338. As will be discussed in further detail below, the channel 2348 may retain a safety

16

arm 124. The semi-circular surface 2346 is vertically above the third surface 2322 and the fourth surface 2324 of the first safety arm holder 2302 relative to the central vertical axis 2309.

The first surface 2340 of the first portion 2336 and the second surface 2344 of the first portion 2336 extend longitudinally relative to the central vertical axis 2309 to a third surface 2350 and a fourth surface 2352 of the second safety arm holder 2304. Furthermore, the first surface 2343 of the second portion 2338 and the second surface 2347 of the second portion 2338 extend to the third surface 2350 and the fourth surface 2352. The third surface 2350 extends to fifth surface 2354 of the second safety arm holder 2304 and the fourth surface 2352 surface extends to a sixth surface 2356 of the second safety arm holder 2304. The fifth surface 2354 and the sixth surface 2356 extend to the semi-circular surface 2346.

A seventh surface 2358 of the second safety arm holder 2304 extends longitudinally relative to the central vertical axis 2309 from the first surface 2332 of the bracket 128 to the third surface 2350. An eighth surface 2360 that is opposite the seventh surface 2358 extends from the first surface 2332 of the bracket 128 to the fourth surface 2352. The first surface 2340, the second surface 2344, the fifth surface 2354, the sixth surface 2356, the seventh surface 2358, and the eighth surface 2360 extend generally perpendicular to the central vertical axis 2309. The third surface 2322 and the fourth surface 2324 extend generally parallel to the second horizontal axis 2308.

FIG. 24 depicts the second side 128B of the bracket 128. The bracket 128 includes a first bracket support holder 2402 and a structurally identical second bracket support holder 2404 that extend longitudinally with respect to the central vertical axis 2309 from a surface 2406 of the bracket 128. Accordingly, for the sake of brevity, only the first bracket support holder 2402 will be discussed in further detail below.

The first bracket support holder 2402 includes a first surface 2408, a second surface 2410, a third surface 2412, and a fourth surface 2414. The first surface 2408 extends at an approximately 45° angle from the surface 2406 to the second surface 2410 and extends between the third surface 2412 and the fourth surface 2414. The second surface 2410 extends at an approximately 45° angle from the surface 2406 to the first surface 2408. The second surface 2410 faces a side of the bracket 128 that is not defined by the first safety arm holder 2302 or the second safety arm holder 2304. The second surface 2410 includes an annular edge 2416 that defines a bracket support receptacle 2418. As will be discussed in further detail below, the bracket support receptacle 2418 may be adapted to receive a bracket support 130. The third surface 2412 and the fourth surface 2414 extend longitudinally relative to the central vertical axis 2309 to the first surface 2408 and the second surface 2410.

FIG. 25A depicts the first safety arm 124a coupled to the bracket 128. In order to couple the safety arm 124a to the bracket 128, the first safety arm 124a is moved in the direction of arrow Q. A safety arm 124 includes a rounded surface 2502 that extends between the first side 902 and the second side 904 of the safety arm 124. A width between opposing sides of the rounded surface 2502 is greater than a width between the first side 902 and the second side 904. Furthermore, the width between opposing sides of the rounded surface 2502 is greater than a width of the opening 2303 and the opening 2307, whereas the width between the first side 902 and the second side 904 is less than a width of the opening 2303 and the opening 2307. Accordingly, when

17

inserting a safety arm **124** into the bracket **128**, the safety arm is oriented so that the first side **902** and the second side **904** are parallel to the central vertical axis **2309** allow a safety arm **124** to nest within the first safety arm holder **2302** or the second safety arm holder **2304**. The safety arm **124** may be rotated in the direction of arrow **S** so that the first side **902** and the second side **904** are perpendicular to the central vertical axis **2309** so that the first safety arm holder **2302** and the second safety arm holder **2304** retain a safety arm **124**.

Furthermore, a safety arm **124** has a midpoint between the first end **360** and the second end **362**. The safety arm **124a** may be moved in either direction of arrow **R** in order to align the midpoint of the safety arm **124a** with a center of the bracket **128**. As depicted in FIG. **25A**, when coupling the safety arm **124a** to the bracket **128**, the light source **345** may be adjacent to the semi-circular surface **2318**.

FIG. **25C** depicts the first safety arm **124a** and the second safety arm **124b** coupled to the bracket **128**. The second safety arm **124b** is coupled to the bracket **128** in an identical manner as the first safety arm **124a**. The safety arm **124b** may be moved until the midpoint of the safety arm **124b** aligns with the midpoint of the first safety arm **124a**. When the first safety arm **124a** and the second safety arm **124b** are coupled to bracket **128** and the midpoints are aligned with a center of the bracket **128**, the first safety arm **124a** and the second safety arm **124b** form an "X" shape.

FIG. **26** depicts the bracket **128** coupled to the bracket support **130**. In this embodiment, the bracket support receptacle **2418** slidably receives the bracket support **130**. The bracket **128**, the bracket support **130**, the first safety arm **124a**, and the second safety arm **124b** form a safety device **2602**. When assembled, the safety device **2602** may stand vertically on a surface **2604**. When standing, the first safety arm **124a**, the second safety arm **124b**, and the bracket support **130** contact the surface **2604** to support the safety device **2602**. While FIG. **26** shows the first end **360** of the first safety arm **124a** and the second safety arm **124b**, contacting the surface **2604**, it is envisioned that the first safety arm **124a** and the second safety arm **124b** may be rotated 180° so that the second end **362** of the first safety arm **124a** and the second safety arm **124b** may contact the surface **2604**.

FIG. **27** depicts the safety device **2602** in operation. When in operation, the safety device **2602** may be positioned adjacent to the vehicle **100**. In one embodiment, when in operation, the safety device **2602** emits light **2702** from the light source **345** of the first safety arm **124a** and the second safety arm **124b**. In another embodiment, when in operation, the safety device **2602** does not emit light.

While FIG. **27** shows light **2702** emitted from only one side of the safety device **2602** it is envisioned, in an embodiment wherein the safety arms **124a** and **124b** include a light source **345** on the first side **902** and the second side **904**, so that the safety device **2602** may emit light **2702** from multiple sides. In this embodiment, the safety device **2602** may illuminate multiple areas.

Various inventive concepts may be embodied as one or more methods, of which an example has been provided. The acts performed as part of the method may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include performing some acts simultaneously, even though shown as sequential acts in illustrative embodiments.

While various inventive embodiments have been described and illustrated herein, those of ordinary skill in the

18

art will readily envision a variety of other means and/or structures for performing the function and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the inventive embodiments described herein. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the inventive teachings is/are used. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific inventive embodiments described herein. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, inventive embodiments may be practiced otherwise than as specifically described and claimed. Inventive embodiments of the present disclosure are directed to each individual feature, system, article, material, kit, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, kits, and/or methods, if such features, systems, articles, materials, kits, and/or methods are not mutually inconsistent, is included within the inventive scope of the present disclosure.

All definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

The articles "a" and "an," as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean "at least one." The phrase "and/or," as used herein in the specification and in the claims (if at all), should be understood to mean "either or both" of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with "and/or" should be construed in the same fashion, i.e., "one or more" of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the "and/or" clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to "A and/or B," when used in conjunction with open-ended language such as "comprising" can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc. As used herein in the specification and in the claims, "or" should be understood to have the same meaning as "and/or" as defined above. For example, when separating items in a list, "or" or "and/or" shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as "only one of" or "exactly one of," or, when used in the claims, "consisting of," will refer to the inclusion of exactly one element of a number or list of elements. In general, the term "or" as used herein shall only be interpreted as indicating exclusive alternatives (i.e. "one or the other but not both") when preceded by terms of exclusivity, such as "either," "one of," "only one of," or "exactly one of." "Consisting essentially of," when used in the claims, shall have its ordinary meaning as used in the field of patent law.

As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

When a feature or element is herein referred to as being “on” another feature or element, it can be directly on the other feature or element or intervening features and/or elements may also be present. In contrast, when a feature or element is referred to as being “directly on” another feature or element, there are no intervening features or elements present. It will also be understood that, when a feature or element is referred to as being “connected”, “attached” or “coupled” to another feature or element, it can be directly connected, attached or coupled to the other feature or element or intervening features or elements may be present. In contrast, when a feature or element is referred to as being “directly connected”, “directly attached” or “directly coupled” to another feature or element, there are no intervening features or elements present. Although described or shown with respect to one embodiment, the features and elements so described or shown can apply to other embodiments. It will also be appreciated by those of skill in the art that references to a structure or feature that is disposed “adjacent” another feature may have portions that overlap or underlie the adjacent feature.

Spatially relative terms, such as “under”, “below”, “lower”, “over”, “upper”, “above”, “behind”, “in front of”, and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if a device in the figures is inverted, elements described as “under” or “beneath” other elements or features would then be oriented “over” the other elements or features. Thus, the exemplary term “under” can encompass both an orientation of over and under. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly. Similarly, the terms “upwardly”, “downwardly”, “vertical”, “horizontal”, “lateral”, “transverse”, “longitudinal”, and the like are used herein for the purpose of explanation only unless specifically indicated otherwise.

Although the terms “first” and “second” may be used herein to describe various features/elements, these features/elements should not be limited by these terms, unless the

context indicates otherwise. These terms may be used to distinguish one feature/element from another feature/element. Thus, a first feature/element discussed herein could be termed a second feature/element, and similarly, a second feature/element discussed herein could be termed a first feature/element without departing from the teachings of the present invention.

An embodiment is an implementation or example of the present disclosure. Reference in the specification to “an embodiment,” “one embodiment,” “some embodiments,” “one particular embodiment,” or “other embodiments,” or the like, means that a particular feature, structure, or characteristic described in connection with the embodiments is included in at least some embodiments, but not necessarily all embodiments, of the invention. The various appearances “an embodiment,” “one embodiment,” “some embodiments,” “one particular embodiment,” or “other embodiments,” or the like, are not necessarily all referring to the same embodiments.

If this specification states a component, feature, structure, or characteristic “may”, “might”, or “could” be included, that particular component, feature, structure, or characteristic is not required to be included. If the specification or claim refers to “a” or “an” element, that does not mean there is only one of the element. If the specification or claims refer to “an additional” element, that does not preclude there being more than one of the additional element.

As used herein in the specification and claims, including as used in the examples and unless otherwise expressly specified, all numbers may be read as if prefaced by the word “about” or “approximately,” even if the term does not expressly appear. The phrase “about” or “approximately” may be used when describing magnitude and/or position to indicate that the value and/or position described is within a reasonable expected range of values and/or positions. For example, a numeric value may have a value that is $\pm 0.1\%$ of the stated value (or range of values), $\pm 1\%$ of the stated value (or range of values), $\pm 2\%$ of the stated value (or range of values), $\pm 5\%$ of the stated value (or range of values), $\pm 10\%$ of the stated value (or range of values), etc. Any numerical range recited herein is intended to include all sub-ranges subsumed therein.

Additionally, any method of performing the present disclosure may occur in a sequence different than those described herein. Accordingly, no sequence of the method should be read as a limitation unless explicitly stated. It is recognizable that performing some of the steps of the method in a different order could achieve a similar result.

In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-closed transitional phrases, respectively, as set forth in the United States Patent Office Manual of Patent Examining Procedures.

In the foregoing description, certain terms have been used for brevity, clarity, and understanding. No unnecessary limitations are to be implied therefrom beyond the requirement of the prior art because such terms are used for descriptive purposes and are intended to be broadly construed.

Moreover, the description and illustration of various embodiments of the disclosure are examples and the disclosure is not limited to the exact details shown or described.

21

What is claimed:

1. A vehicle comprising:
a first safety arm;
a first coupler on the first safety arm and rigidly fixed to the first safety arm, wherein the first coupler couples the first safety arm to the vehicle and wherein the first safety arm is adapted to fall away from the vehicle without damaging the vehicle when an object contacts the first safety arm; and
a receiver connected to the vehicle and coupled to the first coupler, wherein a portion of the first coupler is disposed inside of the receiver.
2. The vehicle of claim 1, further comprising:
a trailer; and
a cab connected to the trailer, wherein the first safety arm is coupled to the trailer.
3. The vehicle of claim 1, further comprising:
a trailer; and
a cab connected to the trailer, wherein the first safety arm is coupled to the cab.
4. The vehicle of claim 1, further comprising:
a second safety arm;
a second coupler on the second safety arm, wherein the second coupler couples the second safety arm to the vehicle and wherein the second safety arm is adapted to fall away from the vehicle without damaging the vehicle when the object contacts the second safety arm.
5. The vehicle of claim 4, further comprising:
a trailer; and
a cab connected to the trailer, wherein the first safety arm is coupled to the trailer and the second safety arm is coupled to the cab.
6. The vehicle of claim 4, further comprising:
a trailer; and
a cab connected to the trailer, wherein the first safety arm and the second safety arm are coupled to the trailer.
7. The vehicle of claim 6, further comprising:
a first side of the trailer;
a second side of the trailer that is opposite the first side; and
a nozzle assembly on the first side, wherein the first safety arm and the second safety arm are coupled to the first side forward and rearward of the nozzle assembly, respectively.
8. The vehicle of claim 7, further comprising:
a hose holder connected to the first side of the trailer adjacent to the nozzle assembly, wherein the first safety arm and the second safety arm are coupled to the hose holder.
9. The vehicle of claim 1, further comprising:
a ferromagnetic surface on one of the coupler and the first safety arm;
a magnet on the other of the coupler and the first safety arm; wherein the magnet couples the first safety arm to the ferromagnetic surface.
10. The vehicle of claim 1, further comprising:
a cab;
a trailer connected to the cab;
a nozzle assembly on the trailer; and
a hose holder connected to the trailer adjacent to the nozzle assembly, wherein the receiver is connected to the hose holder.
11. The vehicle of claim 1, wherein the first coupler is adapted to break without damaging the vehicle.
12. The vehicle of claim 11, wherein the receiver defines a keyhole and the first coupler includes a first portion within

22

the keyhole and a second portion outside of the keyhole, that is adapted to break when the first safety arm contacts the object.

13. The vehicle of claim 1, wherein the first coupler is adapted to fall away from the receiver with the first safety arm without damaging the vehicle.

14. The vehicle of claim 13, wherein the receiver includes a triangular arm receiver and the first coupler rests within the triangular arm receiver.

15. The vehicle of claim 13, wherein the receiver includes a hemispherical surface and the first coupler includes a hemispherical member that abuts the hemispherical surface of the receiver.

16. The vehicle of claim 1, wherein the first safety arm includes a light source.

17. The vehicle of claim 16, wherein the first safety arm includes a battery in electrical communication with the light source.

18. The vehicle of claim 16, further comprising:
a power source; and
a power outlet in electrical communication with the power source and connected to a plug of the first safety arm, wherein the plug is in electrical communication with the light source.

19. The vehicle of claim 1, wherein the first safety arm is coupled to the vehicle without any support structure depending from the first safety arm.

20. A method for detaching a safety arm from a vehicle comprising:

providing a coupler on a safety arm, wherein the coupler is rigidly fixed to the safety arm;
attaching the safety arm to a vehicle with the coupler;
housing a portion of the coupler inside of a receiver that is connected to the vehicle;
applying a force to the safety arm; and
detaching the safety arm from the vehicle with the applied force without damaging the vehicle.

21. The method for detaching a safety arm from a vehicle of claim 20, further comprising:
illuminating an area adjacent to the vehicle with a light source on the arm while the arm is attached to the vehicle.

22. The method for detaching a safety arm from a vehicle of claim 20, further comprising:
moving the vehicle;
contacting an object with the arm while the vehicle is moving, wherein the contact applies the force to the arm; and

detaching the arm from the vehicle while the vehicle is moving without damaging the vehicle.

23. The method for detaching a safety arm from a vehicle of claim 20, further comprising:
breaking a coupler of the arm with the applied force, wherein breaking the coupler detaches the arm from the vehicle without damaging the vehicle.

24. The method for detaching a safety arm from a vehicle of claim 20, wherein the arm detaches from the vehicle by falling away from the vehicle without damaging the vehicle.

25. A vehicle comprising:
a first safety arm;
a first coupler the first safety arm, wherein the first coupler couples the first safety arm to the vehicle and wherein the first safety arm is adapted to fall away from the vehicle without damaging the vehicle when an object contacts the first safety arm;

23

a second safety arm; and
a second coupler on the second safety arm, wherein the
second coupler couples the second safety arm to the
vehicle and wherein the second safety arm is adapted to
fall away from the vehicle without damaging the 5
vehicle when the object contacts the second safety arm.

* * * * *

24